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Experimental and modeling study of the oxidation of *n*- and *iso*-butanal

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ABSTRACT

Understanding the kinetics of large molecular weight aldehydes is essential in the context of both conventional and alternative fuels. For example, they are key intermediates formed during the low-temperature oxidation of hydrocarbons as well as during the high-temperature oxidation of oxygenated fuels such as alcohols. In this study, an experimental and kinetic modeling investigation of *n*-butanal (*n*-butyraldehyde) and *iso*-butanal (*iso*-butyraldehyde or 2-methylpropanal) oxidation kinetics was performed. Experiments were performed in a jet stirred reactor and in counterflow flames over a wide range of equivalence ratios, temperatures, and pressures. The jet stirred reactor was utilized to observe the evolution of stable intermediates and products for the oxidation of *n*- and *iso*-butanal at elevated pressures and low to intermediate temperatures. The counterflow configuration was utilized for the determination of laminar flame speeds. A detailed chemical kinetic interpretative model was developed and validated consisting of 244 species and 1198 reactions derived from a previous study of the oxidation of propanal (propionaldehyde). Extensive reaction pathway and sensitivity analysis was performed to provide detailed insight into the mechanisms governing low-, intermediate-, and high-temperature reactivity. The simulation results using the present model are in good agreement with the experimental laminar flame speeds and well within a factor of two of the speciation data obtained in the jet stirred reactor.

1. Introduction

The role of large molecular weight aldehydes in a variety of important combustion phenomena has been long realized. Attempts of early combustion scientists to better understand and eventually prevent engine knock led to combustion research that focused on the low-temperature or cool-flame phenomena of hydrocarbon fuels. Prior to 1950, Townend [1] and von Elbe and Lewis [2] recognized that the mechanism of low-temperature oxidation of hydrocarbons results in the formation of aldehydes as key intermediates. Experimental results from the 1950s to the 1970s (e.g., [3–5]) utilizing static reactors to probe the low-temperature oxidation regime confirmed the presence of large molecular weight aldehydes such as propanal (propionaldehyde), *n*-butanal (*n*-butyraldehyde), and *iso*-butanal (*iso*-butyraldehyde or 2-methylpropanal). Later, Benson [6] proposed that aldehydes were the main product from the chain decomposition of primary hydroperoxides. Thus, it has been established that aldehydes are key intermediates in the chain branching mechanism governing low-temperature oxidation chemistry of hydrocarbons.

More recently, the oxidation of neat hydrocarbon compounds at low-temperature conditions has been the subject of extensive fundamental numerical and experimental studies (e.g., [7–12]). These experiments, performed in rapid compression machines (e.g., [7–9]), flow reactors (e.g., [10]), and jet stirred reactors (e.g., [11,12]), identified and quantified the presence of C_4 saturated aldehydes (*n*- and *iso*-butanal) as key intermediates formed during the low-temperature oxidation of hydrocarbons.

Among the main concerns regarding the combustion of conventional petroleum derived hydrocarbons in spark ignition (SI) gasoline and compression ignition (CI) diesel engines is the emission of toxic compounds including aldehydes that are classified as mobile source air toxics (MSAT). Aldehydes formed during the oxidation of hydrocarbons are expected to be part of conventional engine emissions. Indeed this has been confirmed in SI and CI engines, in which large molecular weight aldehydes, including *n*- and *iso*-butanal, have been detected in exhaust gases (e.g., [13–22]). Additionally, the presence of *n*- and *iso*-butanal in the lower atmosphere may influence the atmospheric oxidation capacity [23].

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The recent drive to both improve the efficiency of SI and CI engines and reduce emissions of nitrogen oxides (NO_x) and particulate matter (PM), including soot, have led to the development of new in cylinder combustion technologies including the homogenous charged compression ignition (HCCI) and premixed charged compression ignition (PCCI) engines. These advanced engine concepts rely on compression ignition to initiate volumetric combustion, therefore their success relies heavily on the details of low-temperature chemistry [24]. Recent studies have shown that relative to conventional engines (SI or CI) the emissions of NO_x and PM from HCCI and PCCI engines decrease, on the other hand emissions of large molecular weight aldehydes increases (e.g., [25–29]).

Alternative fuels, including large molecular weight bio-derived alcohols, have received notable interest recently in the ongoing effort to achieve partial independence from fossil fuels. The combustion of large molecular weight alcohols, including the isomers of butanol, have been linked to an increase in aldehydes emissions relative to fossil fuels. These alcohols demonstrate minor low-temperature reactivity. Nevertheless, the primary stable intermediate formed after H-abstraction and β-scission of the parent alcohol at high-temperatures can be a saturated aldehyde. At lower temperatures, aldehydes are formed via the reaction of *1*-hydroxyalkyl radicals with O_2. For example, *n*-butanal results from the oxidation of *n*-butanol, and *iso*-butanal is produced by the oxidation of *iso*-butanol. The two saturated C_4 aldehydes have been identified to form as intermediates repeatedly in fundamental experimental and kinetic modeling studies of the oxidation of the butanol isomers (e.g., [30–46]), and they have been detected in the emissions of research engines fueled by butanols (e.g., [47,48]).

Given the important role of large molecular weight aldehydes in combustion, it is somewhat surprising that there are few fundamental experimental or kinetic modeling studies of these compounds. Early work on *n*- and *iso*-butanal has been performed in flow reactors (e.g., [49]), and a number of fuel-specific elementary rate constants have been evaluated (e.g., [50,51]). More recently, da Silva and Bozzelli [52] have determined the enthalpies of formation and bond dissociation energies of a series of straight chain aldehydes, including *n*-butanal. The reaction rate of H with *n*-butanal was computed by Simmie and Curran [53]. Davidson et al. [54] measured ignition delay time data for *n*-butanal/oxygen/argon mixtures over a wide range of equivalence ratios, ϕ's, and estimated rates for the uni-molecular decomposition reactions of *n*-butanal. To the authors best knowledge there are no relevant oxidation studies of *iso*-butanal available in recent literature.

Clearly, improving the oxidation chemistry of large molecular weight aldehydes is essential in better understanding the mechanisms of auto-ignition in advanced engines (e.g., HCCI and PCCI engines). In addition, this understanding will lead to new strategies to better mitigate the emissions of aldehydes from the combustion of petroleum based fuels and bio-derived fuels. To this end, the goal of this study was twofold. First, to provide archival experimental data on the oxidation of *n*- and *iso*-butanal over a wide range of ϕ's, temperatures, and pressures. Second, to use the experimental data as validation targets of an interpretive reaction model for the low- and high-temperature oxidation *n*- and *iso*-butanal. This investigation is the second part of an ongoing study by the authors on the oxidation of the aldehydes having previously addressed the oxidation of propanal [55].

2. Experimental approach

2.1. Jet-stirred reactor

The experimental configuration and approach has been described in earlier studies [56,57]. The jet-stirred reactor (JSR) consists of a 4 cm diameter fused silica sphere (42 cm^3) with four injection nozzles of 1 mm internal diameter (i.d.). Prior to the injectors, the reactants were diluted with nitrogen (<100 ppm H_2O, <50 ppm O_2, <1000 ppm Ar, <5 ppm H_2 from Air Liquide) and mixed. A high degree of dilution was used (0.15% mol. of fuel), limiting temperature gradients and heat release in JSR. The reactants were high-purity oxygen (99.995% pure form Air Liquide) and high-purity fuels (*n*-butanal >99% pure from Aldrich, CAS 123-72-8 and *iso*-butanal >99% pure from Aldrich, CAS 78-84-2). The reactants were preheated before injection to minimize temperature gradients inside the reactor. A Shimadzu HPLC pump (LC10 AD VP) with an on-line degasser (Shimadzu DGU-20 A3) was used to deliver the fuel to an in-house atomizer–vaporizer assembly maintained at 200 °C. Good thermal homogeneity along the vertical axis of the reactor (gradients of ca. 1 K/cm) was observed during the experiments by Pt–Pt/Rh-10% thermocouple measurements (wires of 0.1 mm sited inside a thin-wall fused silica tube). The reacting mixtures were sampled using a movable fused silica low-pressure sonic probe and sent to analyzers via a Teflon heated line maintained at 200 °C. Online analyses were performed using a FTIR spectrometer (10 m path length, resolution of 0.5 cm^{-1}, 200 mBar in the cell). Off-line analyses were performed, after samples were collected and stored in 1 L Pyrex bulbs, using gas chromatographs (GC) equipped with capillary columns (0.32 mm i.d. DB-624, and 0.32 mm i.d. CP-Al2O3-KCl, and 0.53 mm i.d. Carboplot-P7). A thermal conductivity detector (TCD) and a flame ionization detector (FID) were used. Two GC–MS (quadrupole V1200, Varian Inc.) operating with electron ionization (70 eV) were used for the identification of products.

The experiments were performed at steady state as well as at constant pressure and mean residence time, τ. The reactants were continuously flowing in the reactor, while inside the JSR, the temperature of the gases was increased stepwise. A good repeatability of the measurements and a reasonably good carbon balance (better than 100 ± 15%) were obtained in this series of experiments.

2.2. Laminar flame speeds

Laminar flame speeds, S_u^o's, were measured in the counterflow configuration similarly to previous studies (e.g., [37]). Digital Particle Image Velocimetry (DPIV) was utilized to characterize the flowfield (e.g., [58,59]).

In order to determine S_u^o, the minimum point of the velocity profile just upstream of the flame is measured and defined as the reference velocity, $S_{u,\mathrm{ref}}$. The absolute value of the maximum velocity gradient in the hydrodynamic zone is defined as the imposed strain rate, K. By plotting $S_{u,\mathrm{ref}}$ as a function of K, S_u^o could be determined by non-linearly extrapolating $S_{u,\mathrm{ref}}$ to $K = 0$ using a computationally-assisted technique (e.g., [60]). Particular care was taken to maintain a plug-flow profile at the nozzle exit in order to increase the accuracy of the non-linear extrapolations [61].

The liquid fuel vaporization was accomplished by using a quartz Meinhard® nebulizer to inject the fuel as a fine aerosol into a heated glass vaporization chamber. In order to maintain the partial pressure of the fuel below its vapor pressure, the unburned fuel stream temperature, T_u, at the burner exit was maintained at $T_u = 343$ K. All measurements were carried out with nozzles with diameter, $D = 14$ mm. The burner separation distance, L, was equal to D. Overall, the uncertainties in ϕ were determined to be less than 0.5%. The temperature of the fuel stream, measured at the center of the nozzle exit, fluctuated within ±2 K. The sampling errors in $S_{u,\mathrm{ref}}$ and K were determined and their 2σ standard deviations are indicated with uncertainty bars in all figures. More details of the uncertainty analysis can be found in Ref. [62].

3. Numerical approach

S_u^o was modeled using the PREMIX code ([63,64]). The variation of $S_{u,\mathrm{ref}}$ with K was computed using a CHEMKIN based opposed-jet code originally developed by Kee and coworkers [65]. Both codes have been modified to account for thermal radiation of CH_4, CO, CO_2, and H_2O in the optically thin limit [66], and are coupled with the Sandia CHEMKIN [67] and Transport [68] subroutine libraries. The H and H_2 diffusion coefficients of several key pairs are based on a recently updated set of Lenard-Jones parameters [69]. The JSR data were modeled using the perfectly stirred reactor (PSR) code [70].

The proposed kinetic reaction mechanism was built as an extension to that proposed earlier for the oxidation of propanal [55]. It consists of 1198 reversible reactions involving 244 species. For the new species, the thermochemical properties were computed using Benson's group and bond additivity methods [71]. It includes the same types of reactions, i.e., fuel decomposition, reaction with dioxygen, H-atom abstraction by radicals and atoms, and peroxidation of primary fuel radicals at low-temperature ($R + O_2 \rightleftarrows ROO$) followed by peroxy radical isomerization ($ROO \rightleftarrows QOOH$), addition of O_2 to QOOH ($QOOH + O_2 \rightleftarrows OOQOOH$), and isomerization of OOQOOH followed by decomposition reactions ($OOQOOH \rightleftarrows HOOQ'OOH \rightarrow OH + OQ'OOH$ and $OQ'OOH \rightarrow OH + Products$). The methodology utilized to develop the rate constants used in this study are consistent with those used previously for modeling the oxidation of propanal under similar conditions [55].

The pertinent transport parameters were estimated using the correlation of corresponding states of Tee et al. [72] similarly to Wang and Frenklach [73] and Holley et al. [74]. In the correlation of corresponding states, the Lenard-Jones self-collision diameter and well depth are related to the critical pressure and temperature. The transport parameters of fuel radicals were assumed to be equal to those of analogous fuel species [73,74].

4. Results and discussions

4.1. Jet-stirred reactor

JSR experiments for the oxidation of both *n*- and *iso*-butanal were performed at a mean residence time of 0.7 s and at a constant pressure of 10 atm. For all experiments, the fuel concentration in the reactant stream entering the reactor was held constant at 1500 ppm and experiments were carried out for $\phi = 0.30$, 0.50, 1.00, and 2.00. For a fixed equivalence ratio, JSR temperature was varied between 500 and 1200 K. The mole fraction of reactants (*n*-butanal, *iso*-butanal, and O_2) and stable intermediates and products (H_2O, CO, CO_2, formaldehyde (CH_2O), acetaldehyde (CH_3CHO), acetone (CH_3COCH_3), 2-propenal (acrolein or C_2H_3CHO), propanal, CH_4, C_2H_2, C_2H_4, C_2H_6, C_3H_6, C_3H_8, and H_2) were quantified using FTIR spectrosopy and GC/MS.

4.1.1. n-Butanal oxidation

Figures 1–4 depict selected experimental results (symbols) and numerical calculations (lines) for the oxidation of *n*-butanal at ϕ = 0.30, 0.50, 1.00, and 2.00 respectively. In each figure, the mole fractions of the oxidizing reactants and the subsequently forming intermediates and products are presented as functions of the fixed reactor temperature. From the experimentally measured fuel

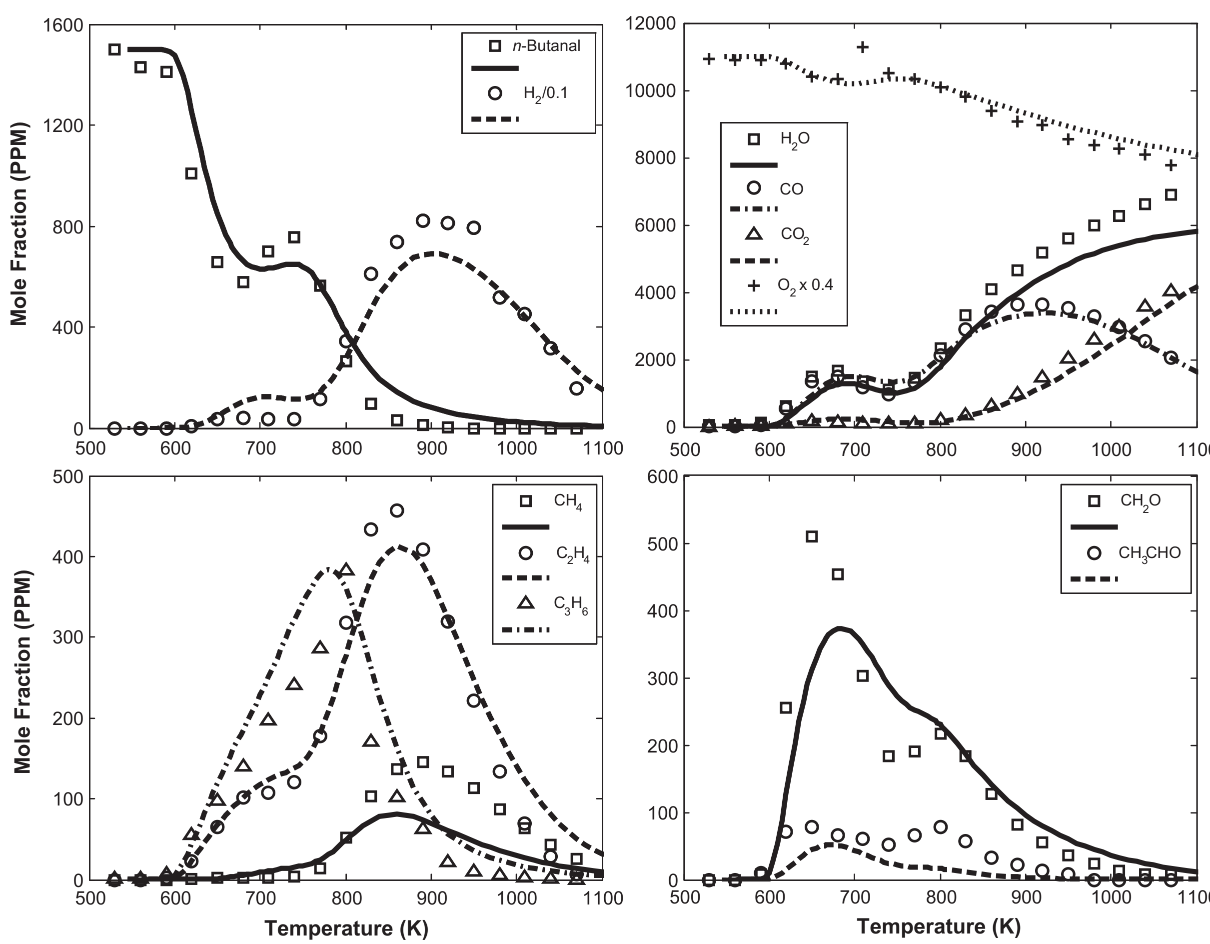


Fig. 1. Experimental (symbols) and computed (lines) mole fraction profiles obtained from the oxidation of *n*-butanal in a JSR at $\varphi = 0.3$, $p = 10$ atm, and $\tau = 0.7$ s.

concentrations, the onset of the cool flame, the negative temperature coefficient (NTC), and the high-temperature kinetic regimes can be distinguished clearly. Little reactivity is observed until the reactor temperature reaches ∼590 K. Between 590 K and 650 K the concentration of *n*-butanal decreases notably indicating the onset of the cool flame or low-temperature reactivity. Between ∼650 K and ∼700 K there is an increase in the reactant concentration as a function of temperature signifying the end of low-temperature reactivity and the onset of NTC behavior. Finally, the concentration of *n*-butanal decreases monotonically with temperature for $T > 700$ K, indicating the transition to the high-temperature kinetic regime.

At the onset of the low-temperature reactivity, experimental results indicate that *n*-butanal consumption is followed by the rapid formation of H_2O, CO, CH_2O, and CH_3CHO. The concentrations of each of these species reach a local maximum between 590 K and 650 K, and CH_2O and CH_3CHO constitute markers of low-temperature reactivity. Within this temperature regime, little conversion of CO to CO_2 was observed. The hydrocarbon intermediates C_3H_6 and C_3H_8 are produced in measurable quantities. As expected, the extent of the low-temperature reactivity was found to be a function of ϕ. As ϕ increases, the *n*-butanal consumption and the attendant intermediates formation decreases. The difference in peak concentration of CH_2O at $\phi = 2.00$ and 0.30 is 50%. Additionally, as φ increases, the temperature range over which low-temperature and NTC behaviors are observed, narrows.

For all ϕ's considered and for $T > 710$ K, *n*-butanal is rapidly consumed. Consumption of reactant is accompanied closely by the rapid formation of the oxygenated intermediates propanal, C_2H_3CHO, CH_2O, and CH_3CHO. The major hydrocarbon intermediates formed included C_3H_6, C_2H_4, and CH_4 which have peak concentrations at ∼800 K, 850 K, and 950 K respectively. The temperature at which CH_4 and CH_3CHO concentrations peak shifts to higher reactor temperatures as ϕ increases. C_2H_4 consumption is followed closely by the formation of C_2H_2. Peak C_2H_4 concentration increases and peak C_3H_6 concentration decrease as ϕ increases.

In Figs. 1–4 the experimental and computed mole fractions using the current interpretive kinetic model are compared. It is observed that the model reproduces the three distinct kinetic regimes as observed in the experiments; low-temperature reactivity, NTC behavior, and second stage ignition. More significantly, the model reproduces well the temperatures at which each regime transitions into the next one. The reduction in the extent of low-temperature reactivity with increasing ϕ and the narrowing of the temperature range over which this reactivity is observed is reproduced by the model. Model predictions and experimental results for the rates of consumption of reactants (*n*-butanal and O_2), and formation major products (H_2O, CO_2, H_2, and CO) are in excellent agreement as well. The predicted concentrations of C_2H_4 and C_3H_6 agree well with the data for fuel-lean conditions. For $\phi = 1.0$ and 2.0 the model underpredicts the peak C_2H_4 concentration. Additionally, for all ϕ's considered, CH_4 production is underpredicted. One notable deficiency in the present model is the underprediction of the extent of NTC behavior. As a result, between ∼650 K and ∼700 K, the concentration of fuel is underpredicted and the concentration of

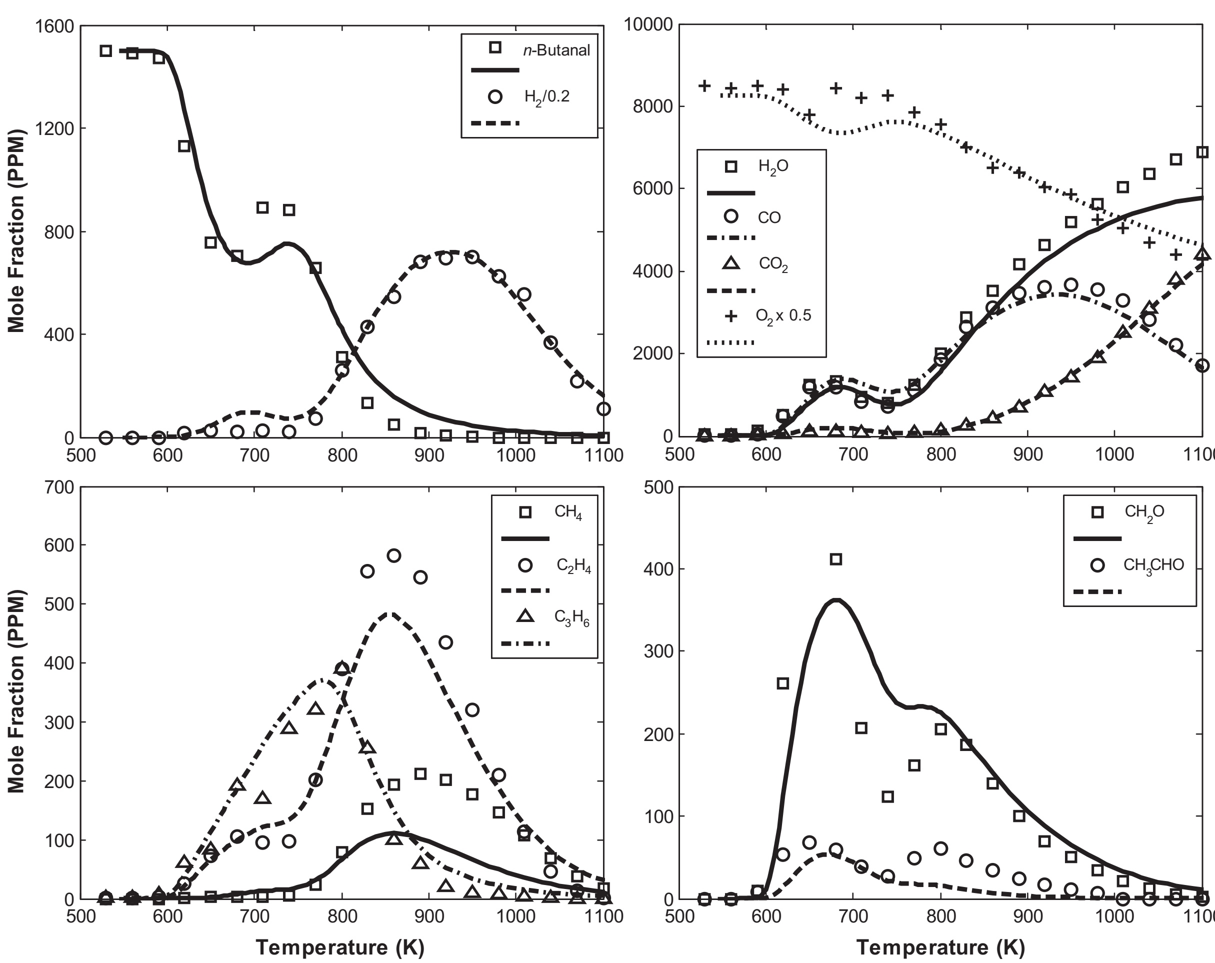


Fig. 2. Experimental (symbols) and computed (lines) mole fraction profiles obtained from the oxidation of *n*-butanal in a JSR at $\varphi = 0.5$, $p = 10$ atm, and $\tau = 0.7$ s.

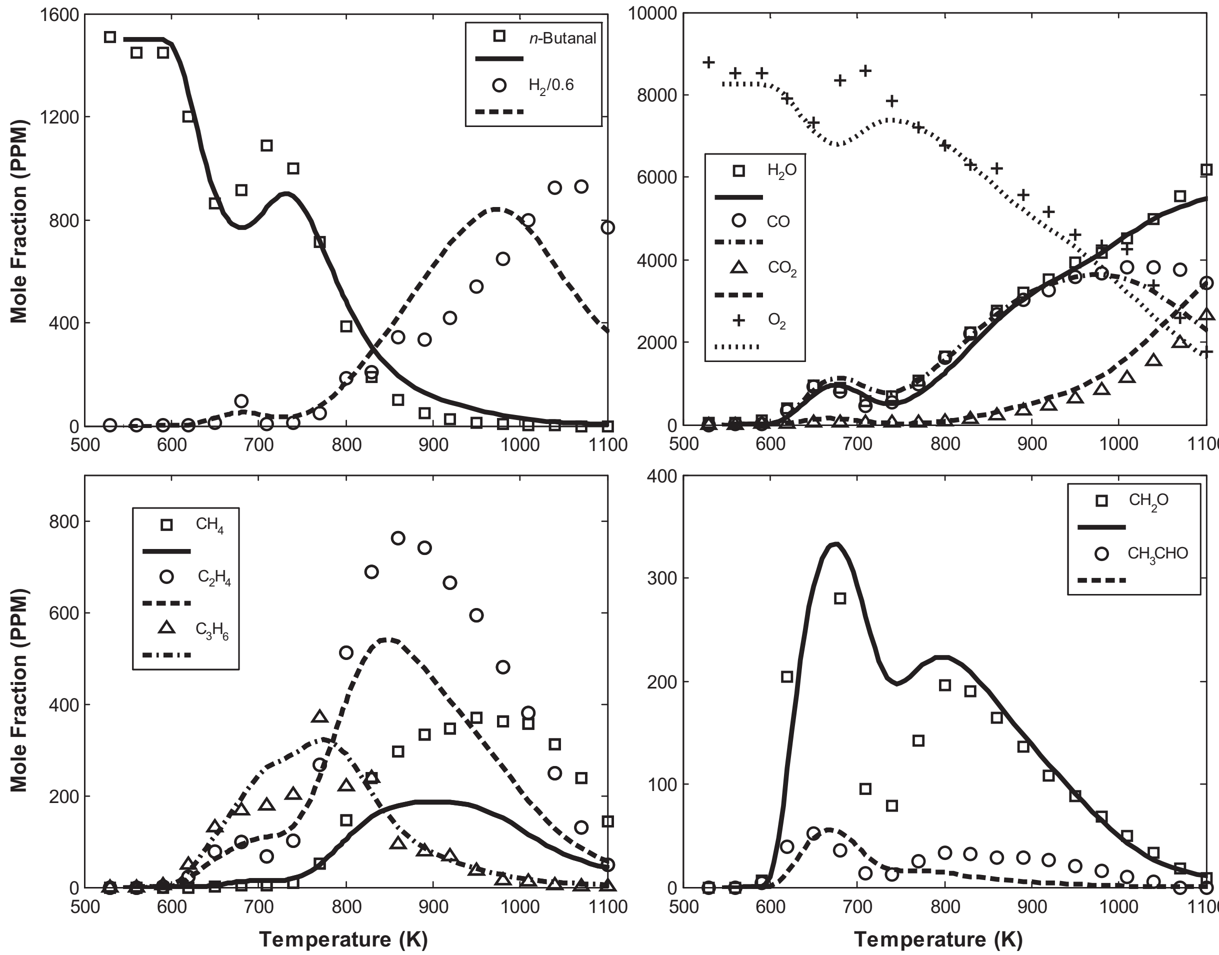


Fig. 3. Experimental (symbols) and computed (lines) mole fraction profiles obtained from the oxidation of *n*-butanal in a JSR at φ = 1.0, *p* = 10 atm, and τ = 0.7 s.

CH_2O overpredicted. There is good reproduction of CH_3CHO concentration at low-temperature and NTC regimes, but the simulations do not reproduce the observed maximum in the high-temperature kinetics regime.

An indication of any kinetic model's performance is its ability to reproduce the relative quantities of intermediates observed in experiments. For example, the model reproduces the experimental ratio of C_2H_4 to C_3H_6. Although discrepancies exist in the predictions of some intermediates, it can be stated with confidence that the present model is capturing to a large extent the details of the oxidation of *n*-butanal.

In order to provide further insight into the controlling reactions and dominant pathways over the low-temperature or cool flame kinetic regime, reaction flux (Fig. 5) and sensitivity analyses (Fig. 6) were performed. The structure and nomenclature of selected species relevant to the following discussion are shown in Table 1. The sensitivity analysis depicted in Fig. 6 is the first order sensitivity of *n*-butanal concentration to the pre-exponential factors of the elementary rate parameters. In Fig. 6, a reaction with negative sensitivity indicates that increasing its rate will result in a reduction in fuel concentration (or increased reactivity) and vice versa. The analysis was performed for a fuel-lean (ϕ = 0.30) *n*-butanal and oxidizer mixture at a reactor temperature of 645 K. Fuel-lean conditions were chosen for this study as they exhibit the highest rates of low-temperature reactivity observed both experimentally and in numerical simulations. At these conditions ~40% of *n*-butanal has been oxidized.

At the onset of low-temperature reactivity, initiation reactions are dominated by H-abstraction. Approximately 80% of fuel consumption occurs through H-abstraction by OH and the remaining 20% through H-abstraction by CH_3O_2 and HO_2. Using the present model, branching ratios of 65:15:15:5 for *n*-C_3H_7CO, C_3H_6CHO-2, C_3H_6CHO-3, and C_3H_6CHO-4 are predicted respectively. These relative branching ratios between the four distinct fuel specific radicals produced following initiation reactions are somewhat expected as the H- on the formyl group is the weakest bond and the CH_2–H bond the strongest within *n*-butanal. The authors have reported similar branching ratios for the oxidation of propanal within the regime of low-temperature reactivity [55].

The results of the sensitivity analysis indicate that increasing the rate of the H-abstraction reaction from the formyl group

$$n\text{-}C_3H_7CHO + OH \leftrightarrow n\text{-}C_3H_7CO + H_2O \quad (R1)$$

has a negative effect on overall reactivity. This is somewhat counterintuitive at first glance since 40% of total *n*-butanal consumption is via R1. *n*-C_3H_7CO subsequently undergoes either β-scission or O_2 addition through

$$n\text{-}C_3H_7CO \leftrightarrow n\text{-}C_3H_7 + CO \quad (R2)$$

$$n\text{-}C_3H_7CO + O_2 \leftrightarrow n\text{-}C_3H_7CO_3. \quad (R3)$$

Using the present model, it is predicted that at 645 K the ratio of *n*-C_3H_7CO consumption via R2 to R3 is 3:7. The majority of

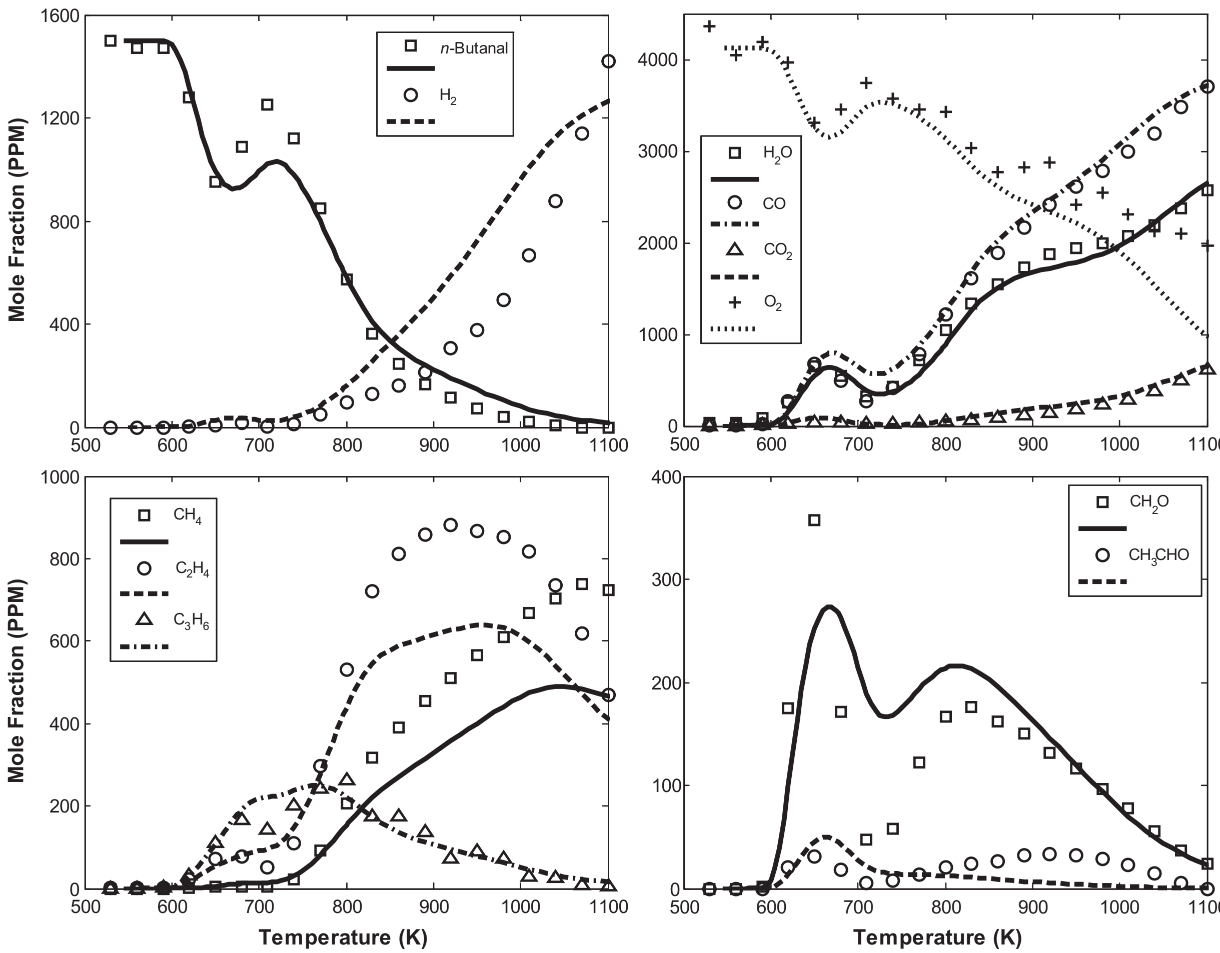


Fig. 4. Experimental (symbols) and computed (lines) mole fraction profiles obtained from the oxidation of *n*-butanal in a JSR at φ = 2.0, *p* = 10 atm, and τ = 0.7 s.

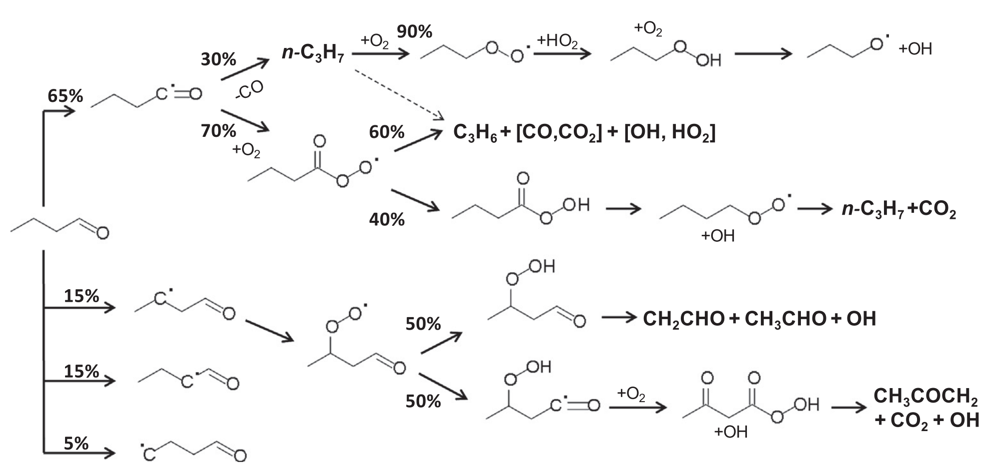


Fig. 5. Major reaction pathways in the case of the lean (φ = 0.3) oxidation of *n*-butanal in a JSR at *p* = 10 atm, τ = 0.7 s, and *T* = 645 K.

subsequent reaction pathways for n-$C_3H_7CO_3$, resulting from R3, produce C_3H_6 without undergoing chain branching type reactions required to sustain or promote low-temperature reactivity. C_3H_6 consumption primarily produces a-C_3H_5 further removing radicals from the system. As a result, an increase in the rate parameter for R1 has the net effect of decreasing low-temperature reactivity observed in numerical calculations.

Reaction pathways for the normal propyl (n-C_3H_7) radical, resulting from R2, are dominated by O_2 addition forming n-$C_3H_7O_2$ which subsequently reacts with HO_2 forming *n*-propyl hydroperoxide (n-$C_3H_7O_2H$). About 10% of n-C_3H_7 undergoes direct elimination leading to propene via,

$$n\text{-}C_3H_7 + O_2 \leftrightarrow C_3H_6 + HO_2. \tag{R4}$$

R4 is responsible for the majority of the C_3H_6 produced within this system. n-$C_3H_7O_2H$ decomposes through degenerate branching and β-scission forming C_2H_5 and CH_2O via

$$n\text{-}C_3H_7O_2H \leftrightarrow n\text{-}C_3H_7O + OH \quad (R5)$$

$$n\text{-}C_3H_7O \leftrightarrow C_2H_5 + CH_2O. \quad (R6)$$

C_2H_5 undergoes an identical sequence of reactions to n-C_3H_7 resulting in $C_2H_5O_2H$, which is also consumed via

$$C_2H_5O_2H \leftrightarrow CH_3CH_2O + OH. \quad (R7)$$

Reactions R5 and R7 provide additional branching steps to the oxidation of n-butanal within the low-temperature reactivity regime hence the strong sensitivity to these two reactions observed in the sensitivity spectrum, Fig. 6.

There is strong negative sensitivity to the H-abstraction reactions resulting in the remaining three fuel specific radicals, C_3H_6CHO-2, C_3H_6CHO-3, and C_3H_6CHO-4. As the present model predicts a very similar reaction sequence for C_3H_6CHO-2, C_3H_6CHO-3, and C_3H_6CHO-4, only the reaction pathways for C_3H_6CHO-3 are shown in Fig. 5 and will be discussed in detail. The first step in the reaction sequence for C_3H_6CHO-3 is O_2 addition resulting in peroxy-aldehyde (C_3H_6CHO-$3O_2$) that is consumed via

$$C_3H_6CHO{-}3O_2 + (HO_2, H_2O_2) \leftrightarrow \text{BUT3OOH1}^{*}\text{O} + (O_2, HO_2) \quad (R8)$$

$$C_3H_6CHO{-}3O_2 \leftrightarrow \text{PRCO3O2H1J}. \quad (R9)$$

R8 results in the formation of the hydroperoxy aldehyde (BUT3OOH1*O). In the second pathway, R9, C_3H_6CHO-$3O_2$ undergoes internal H-abstraction from the formyl group resulting in PRCO3O2H1J. Subsequent reactions of PRCO3O2H1J lead to the formation of the keto-hydroperoxy acid (PRCO3*O1O2H). Both BUT3OOH1*O and PRCO3*O1O2H decompose through degenerate chain branching reactions enhancing the overall low-temperature reactivity. The reaction sequence of C_3H_6CHO-2, C_3H_6CHO-3, and C_3H_6CHO-4, results in significantly more chain branching relative to the decomposition sequence of n-C_3H_7CO.

The results of Fig. 6 show also that there is a notable positive sensitivity (reduces reactivity) to the production of HCO via

$$CH_2O + OH \leftrightarrow HCO + H_2O. \quad (R10)$$

R10 consumes the highly reactive OH to eventually produce the relatively less reactive HO_2 via

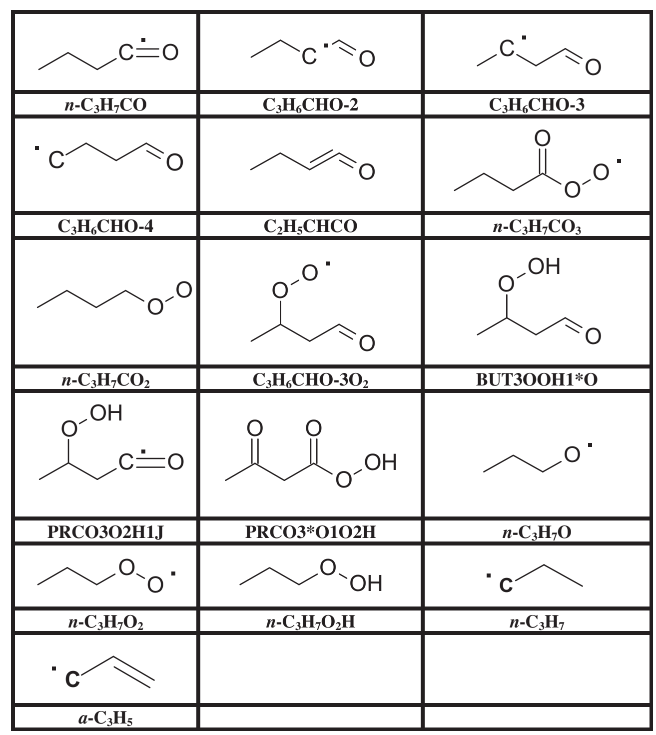
$$HCO + O_2 \leftrightarrow HO_2 + CO. \quad (R11)$$

In order to better understand the changes in the chemical kinetic mechanism of n-butanal oxidation at second stage ignition, sensitivity analysis of the fuel concentration to kinetics was performed and the results are shown in Fig. 7. The sensitivity analysis was performed for a ϕ = 1.0 n-butanal/oxidizer mixture at a reactor temperature of 800 K at which point 60% of initial fuel has been consumed. At these conditions, the formation of n-C_3H_7CO radicals through R12 promotes the overall reactivity, a distinct difference from its effect on low-temperature reactivity shown in Fig. 6,

$$n\text{-}C_3H_7CHO + HO_2 \leftrightarrow n\text{-}C_3H_7CO + H_2O_2. \quad (R12)$$

Within this temperature regime, there is little or no sensitivity to fuel unimolecular decomposition reactions. There is large negative sensitivity to the degenerate branching (R13) and positive sensitivity to the chain terminating (R14) reactions

Table 1
Structure and nomenclature of selected species relevant to the oxidation of n-butanal.



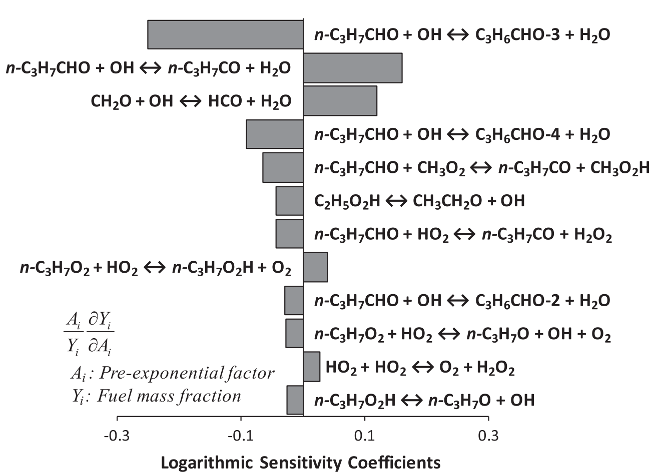


Fig. 6. First-order sensitivity analysis of the oxidation of n-butanal at T = 645 K, φ = 0.3, τ = 0.7 s, p = 10 atm, and approximately 40% total fuel conversion.

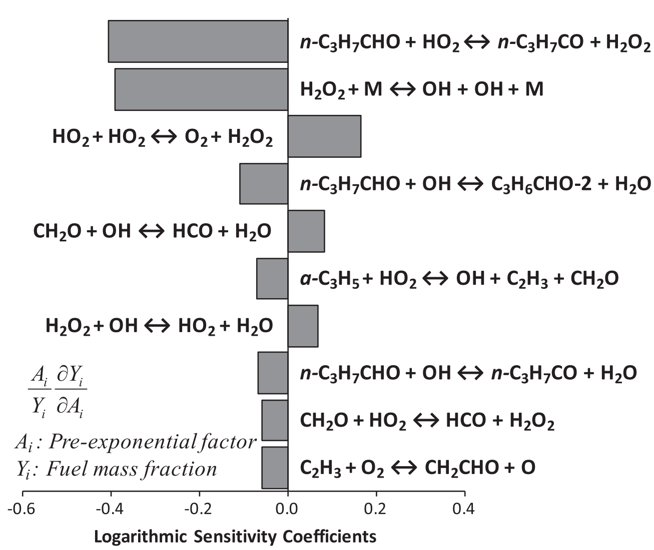


Fig. 7. First-order sensitivity analysis of the oxidation of n-butanal at T = 800 K, φ = 1.0, τ = 0.7 s, and p = 10 atm, at approximately 60% of the total fuel conversion.

$$H_2O_2 + M \leftrightarrow OH + OH + M \quad (R13)$$
$$HO_2 + HO_2 \leftrightarrow H_2O_2 + O_2. \quad (R14)$$

The degenerate branching of H_2O_2 is responsible for driving the system into the second ignition. A detailed reaction flux analysis was performed subsequently at similar conditions to Fig. 7 to provide an overview of the governing consumption pathways. It is interesting to note that there is no significant change in the branching ratios of the initial fuel specific radical formation compared to the low-temperature conditions. As expected, pathways involving O_2 addition consume a significantly smaller fraction of hydrocarbon and oxygenated radical species in favor of β-scission decomposition reactions. For example, the ratio of the consumption of n-C_3H_7CO, R2:R3, shifts from 3:7 (at 645 K) to 9:1 (at 800 K). Similarly, a much larger fraction of n-C_3H_7 now undergoes direct elimination forming C_3H_6 (R4) relative to the formation of n-$C_3H_7O_2H$.

In Fig. 8, the computed overall reactivities for stoichiometric propanal and n-butanal oxidation in a JSR are compared at $\tau = 0.7$ s and $p = 10$ atm. The propanal sub-model was developed and validated recently by the authors [55]. The relative reactivity of these two fuels can be inferred from the comparison of the fuel consumption rates for these two cases. From 600 K to 750 K the n-butanal/oxidizer mixture has a significantly larger extent of low-temperature reactivity compared to the propanal/oxidizer mixture. Additionally, a much smaller fraction of fuel is recovered in the n-butanal case over the NTC regime prior to the onset of high-temperature reactivity. For temperatures larger than 800 K, the overall reactivities tend to converge for both systems. The fact that n-butanal has a significantly larger extent of low-temperature reactivity is to be expected as it has the longer n-alkane chain and is therefore more susceptible to O_2 addition producing RO_2 intermediates. It was observed in the previous study by the authors [55], the propanal radical undergoes largely β-scission to form C_2H_5 and CO whereas the n-butanal radical is more susceptible to O_2 addition. Finally, the overall longer chain length of n-butanal

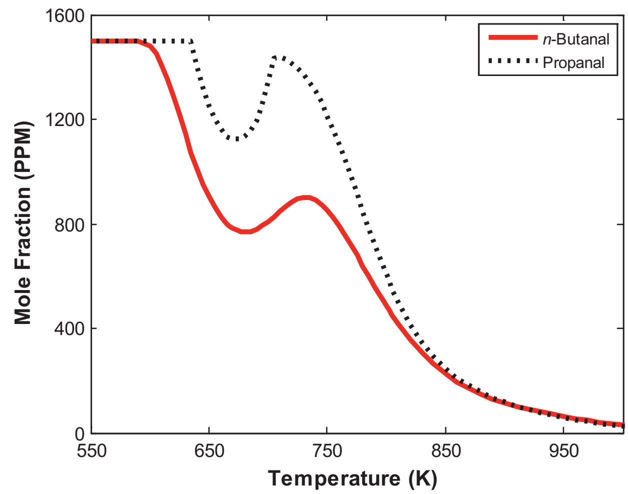


Fig. 8. Comparison of n-butanal and propanal concentration profiles at $\tau = 0.7$ s, $p = 10$ atm, and $\phi = 1.0$.

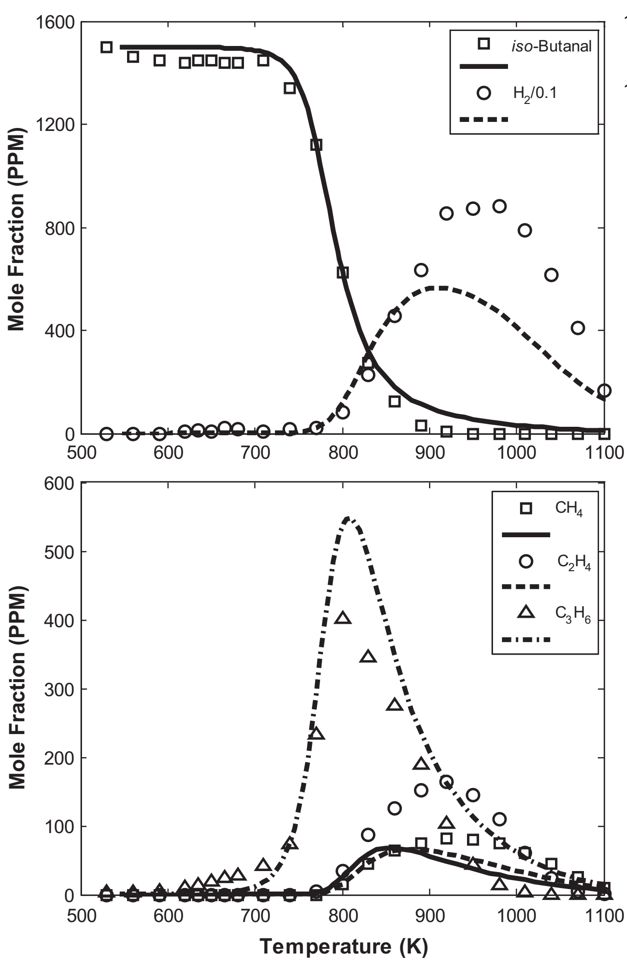


Fig. 9. Experimental (symbols) and computed (lines) mole fraction profiles obtained from the oxidation of iso-butanal in a JSR at $\varphi = 0.3$, $p = 10$ atm, and $\tau = 0.7$ s.

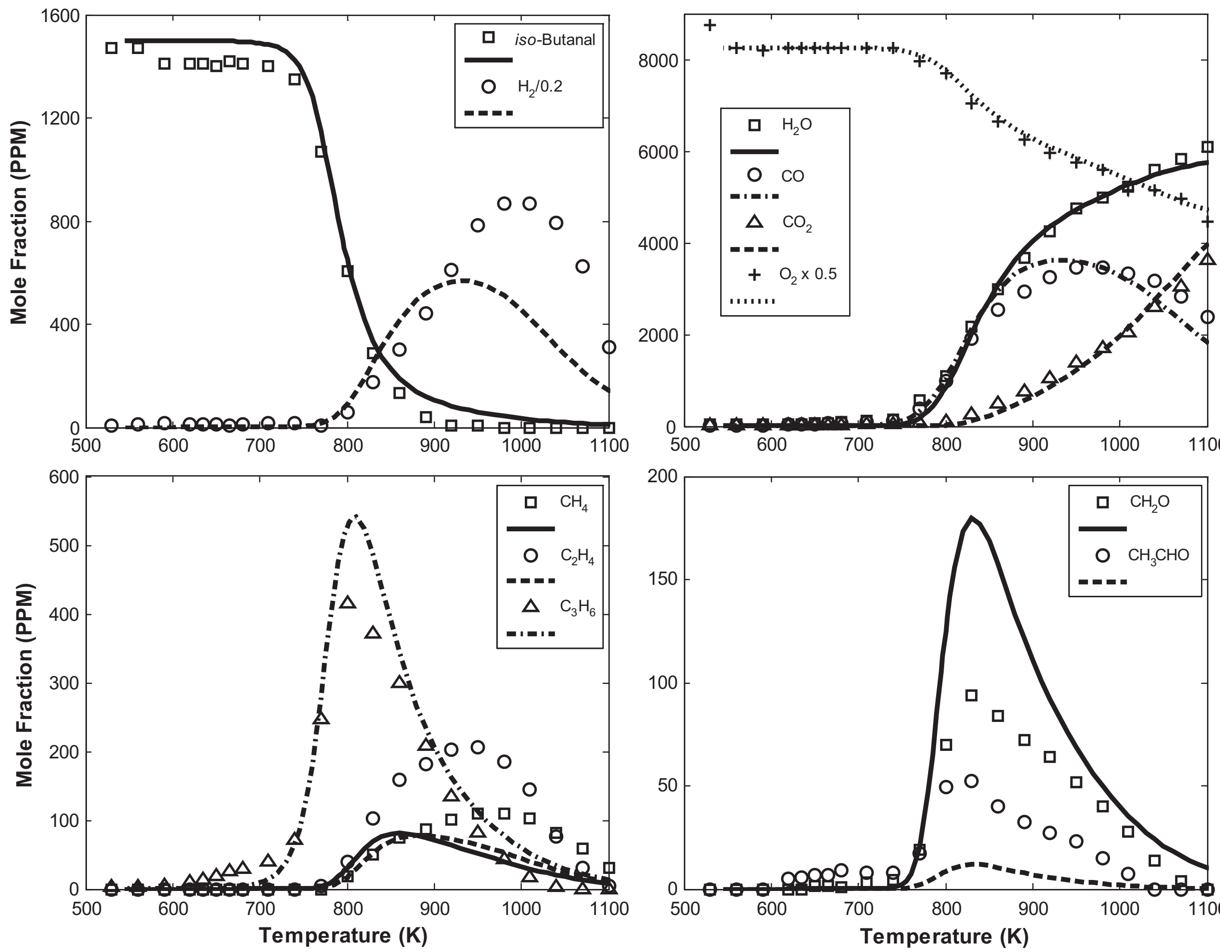


Fig. 10. Experimental (symbols) and computed (lines) mole fraction profiles obtained from the oxidation of *iso*-butanal in a JSR at φ = 0.5, p = 10 atm, and τ = 0.7 s.

leads to more energetically favored pathways for the ROO → QOOH isomerization [24], again increasing low-temperature reactivity.

Salooja [49] compared the reactivity of propanal and *n*-butanal within the cool flame regime, between 473 K and 773 K in a flow reactor type experiment. Contrary to present results Salooja [49] demonstrated that propanal has a higher reactivity relative to *n*-butanal and demonstrated that reactivity decreases with increasing molecular weight or alkane chain length. The results from Ref. [49] are presented in a manner that is somewhat difficult to clearly interpret. As usual, additional experimental testing might clarify this point but present experimental results and numerical calculations support strongly the present conclusion that propanal is far less reactive than *n*-butanal over the cool flame or low-temperature kinetically controlled regime.

4.1.2. iso-Butanal oxidation

Figures 9–12 depict selected experimental data (symbols) and computed results (lines) for the oxidation of *iso*-butanal at ϕ = 0.30, 0.50, 1.00, and 2.00 respectively. The consumption of *iso*-butanal occurs only at $T > \sim$740 K. Between 500 K and 740 K there was no observable reactivity, negligible intermediate formation or fuel consumption, even at the most reactive condition, ϕ = 0.30 shown in Fig. 9. The *iso*-butanal consumption is accompanied with a rapid formation of C_3H_6, CH_3CHO, and CH_2O. These three intermediates have peak concentrations at $\sim$800 K. As C_3H_6 is consumed, the concentrations of C_2H_4 and CH_4 increase. Peak C_2H_4 and CH_4 concentrations are observed to occur at approximately 1000 K. As C_2H_4 is consumed, similar to the observation made for *n*-butanal oxidation, C_2H_2 forms. However, the peak concentration of C_2H_2 and its subsequent consumption was not captured within the experimental temperature range considered.

Comparing the experimentally observed C_2H_4 and C_3H_6 concentration profiles as a function of temperature for the case of *n*- and *iso*-butanal oxidation, there is a clear difference. In the case of *n*-butanal oxidation, C_2H_4 and C_3H_6 are produced concurrently as fuel is consumed. In the case of *iso*-butanal, there is a distinct experimentally observed offset in temperature between the observation of C_3H_6 formation and the onset of C_2H_4 formation with the latter being produced at higher temperatures.

Predictions using the present model for the oxidation of *iso*-butanal in a JSR are compared against the experimental results in Figs. 9–12. Regarding the major products, after the onset of hot-ignition, the predicted values are in very good agreement with experimental results. The model captures closely the concentration profiles for fuel and O_2 consumption along with H_2O, CO, and CO_2 production at all ϕ's considered. H_2 concentration is underpredicted for $T > 900$ K at ϕ = 0.30, 0.50, and 1.00. At ϕ = 2.0, the model overpredicts H_2 concentration. The peak CH_2O concentration predicted by the model is approximately twice compared to its experimental value. Between fuel-lean to stoichiometric conditions, CH_3CHO concentration is underpredicted, however at fuel-rich conditions there is good agreement between predictions and data. Clearly the present model has a deficiency is its ability to reproduce the correct concentration ratio of these two aldehydes.

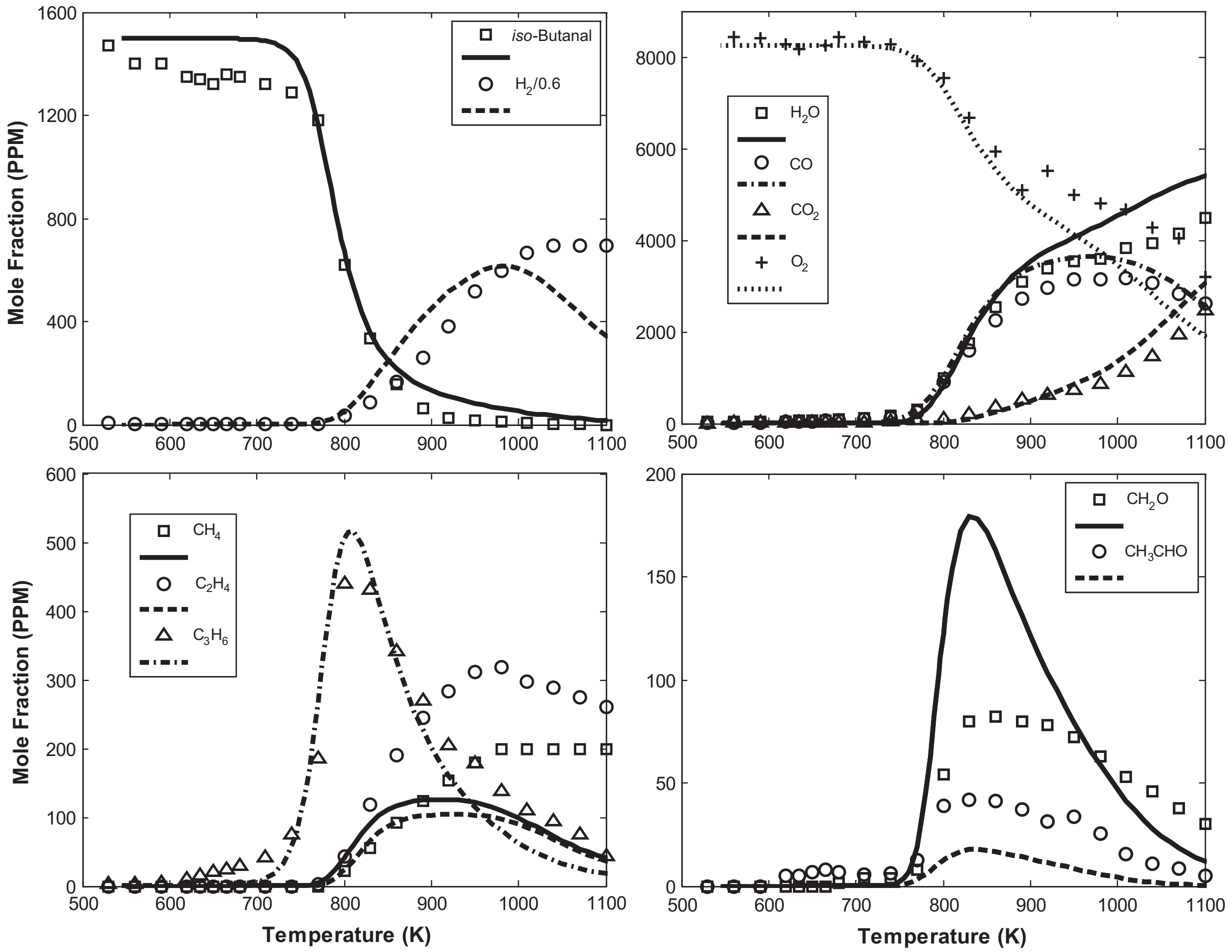


Fig. 11. Experimental (symbols) and computed (lines) mole fraction profiles obtained from the oxidation of *iso*-butanal in a JSR at φ = 1.0, p = 10 atm, and τ = 0.7 s.

Considering next the major hydrocarbon intermediates, only C_3H_6 is correctly predicted at all ϕ's. CH_4 is predicted well only at fuel-rich conditions, while the peak C_2H_4 concentration is underpredicted.

Although the present model for *iso*-butanal oxidation under these particular conditions is unable to produce the peak concentrations for some intermediates, it is still able to capture the majority of experimentally observed trends. For example, the model correctly reproduces the inception temperature for all intermediates. The model captures the 100 K difference between the initial formation of C_3H_6 and C_2H_4. This, along with the accurate prediction of major products and reactants, tends to indicate that it reproduces the oxidation of *iso*-butanal to a satisfactory degree.

Given the notable uncertainties associated with the present oxidation model for *iso*-butanal, sensitivity and reaction path flux analyses were performed to identify the important oxidation pathways, and the results are shown in Figs. 13 and 14 respectively. These analyses were performed at ϕ = 1.0 and T = 800 K, conditions at which approximately 60% of total initial fuel has been reacted. Table 2 depicts the structure and nomenclature of selected species pertinent to the following discussions. Clearly, the present model is missing key reaction sequences or pathways for *iso*-butanal oxidation at the conditions considered in the JSR due to its inability to predict a number of key intermediates profiles correctly. As such the following analysis should be considered as qualitative at best.

Initial fuel decomposition is dominated by H-abstraction reactions. Abstraction by OH and HO_2 accounts for approximately 60% and 30% of total fuel consumption, respectively. The remaining 10% of the fuel initially reacts with H, O, and CH_3. Following initiation, the resulting branching ratios for fuel specific radicals are 75:15:15 for i-C_3H_7CO, t-C_3H_6CHO, and i-C_3H_6CHO respectively. Consistent with analysis performed previously for the oxidation of propanal and n-butanal, H-abstraction from the formyl group is the most significant initiation pathway. This is to be expected as H is more strongly bound at other sites in the *iso*-butanal molecule. i-C_3H_7CO decomposes quickly through unimolecular reactions forming i-C_3H_7 and CO. i-C_3H_7 reacts with O_2 via

$$i\text{-}C_3H_7 + O_2 \leftrightarrow C_3H_6 + HO_2 \quad \text{(R15)}$$

$$i\text{-}C_3H_7 + O_2 \leftrightarrow i\text{-}C_3H_7O_2. \quad \text{(R16)}$$

The reaction rate ratio of RO_2 (R16) formation to the elimination reaction forming propene (R16) for the *iso*-propyl radical is identical to the branching ratio for the similar pair of reactions for n-C_3H_7. i-$C_3H_7O_2$ reacts subsequently with HO_2 forming *iso*-propyl hydroperoxide which undergoes degenerate branching.

Analysis was performed to understand better the failure of the present model to predict correctly the ratio of CH_2O to CH_3CHO concentrations for the oxidation of *iso*-butanal at the conditions considered in the JSR. The largest fraction of CH_2O is produced via the following reactions

$$CH_3CCH_2 + O_2 \leftrightarrow CH_3CO + CH_2O \quad \text{(R17)}$$

$$a\text{-}C_3H_5 + HO_2 \leftrightarrow OH + C_2H_3 + CH_2O. \quad \text{(R18)}$$

Similarly, the majority of CH_3CHO is produced via the following two reactions

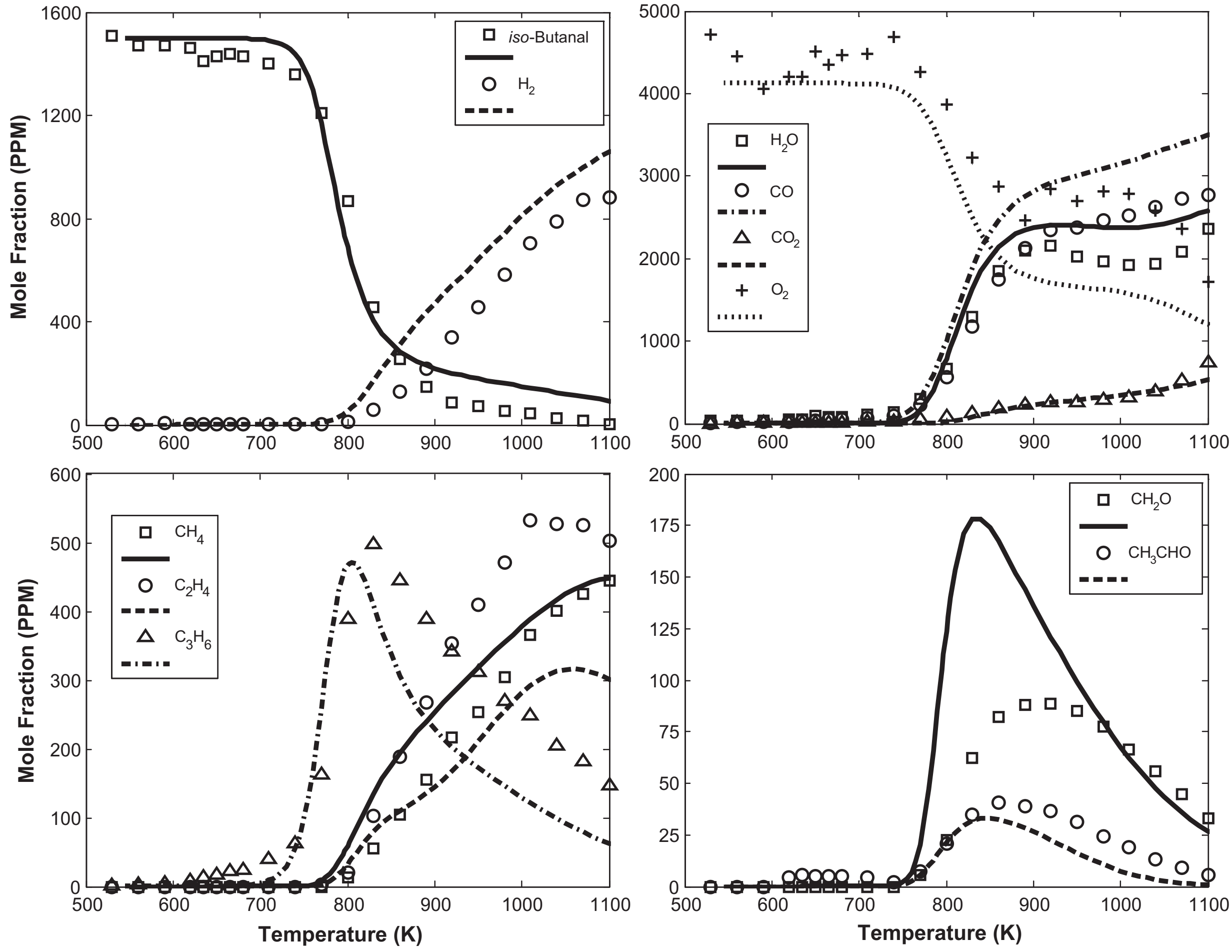


Fig. 12. Experimental (symbols) and computed (lines) mole fraction profiles obtained from the oxidation of *iso*-butanal in a JSR at φ = 2.0, *p* = 10 atm, and τ = 0.7 s.

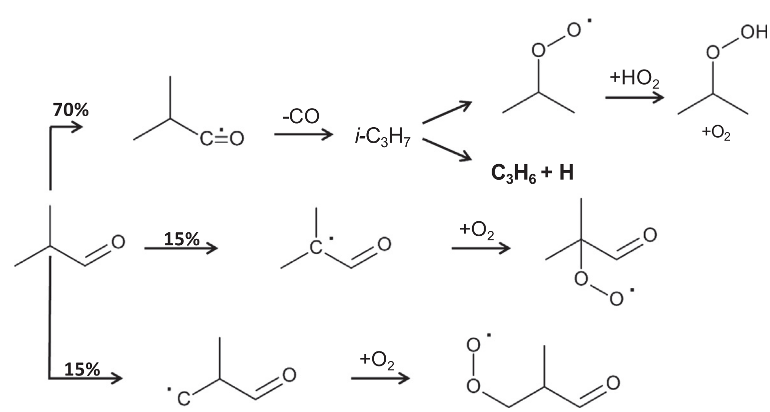


Fig. 13. Major reaction pathways during *iso*-butanal oxidation in a JSR under stoichiometric (φ = 1.0) conditions, at *p* = 10 atm, τ = 0.7 s, and *T* = 800 K.

$$i\text{-}C_3H_7 + HO_2 \leftrightarrow CH_3 + OH + CH_3CHO \quad \text{(R19)}$$

$$CH_3CHCH + O_2 \leftrightarrow HCO + CH_3CHO. \quad \text{(R20)}$$

Correct model predictions of the stable intermediates, CH_2O and CH_3CHO, is heavily reliant on a correct distribution (or ratio) of the radical isomers of C_3H_5 (a-C_3H_5, CH_3CCH_2, and CH_3CHCH). a-C_3H_5, CH_3CCH_2 and CH_3CHCH are produced via the reaction of C_3H_6 and OH. If the branching ratio for the reaction of C_3H_6 with OH leading to the formation of either a-C_3H_5, CH_3CCH_2, or CH_3CHCH is incorrect over the present temperature range considered, this will result in the incorrect model predictions for the distribution of CH_2O and CH_3CHO.

Figure 15 compares the evolution of the *n*-butanal and *iso*-butanal with the reactor temperature at the same conditions shown in Figs. 1 and 11. Clearly *iso*-butanal has no low-temperature reactivity which is as expected since CH_3 substitution (i.e., branching) in the molecular structure has long been established to reduce low-temperature reactivity. The reactivities of both fuels converge after around 800 K. Both *n*-butanal and *iso*-butanal have nearly identical temperatures of the onset of high-temperature reactivity. In this

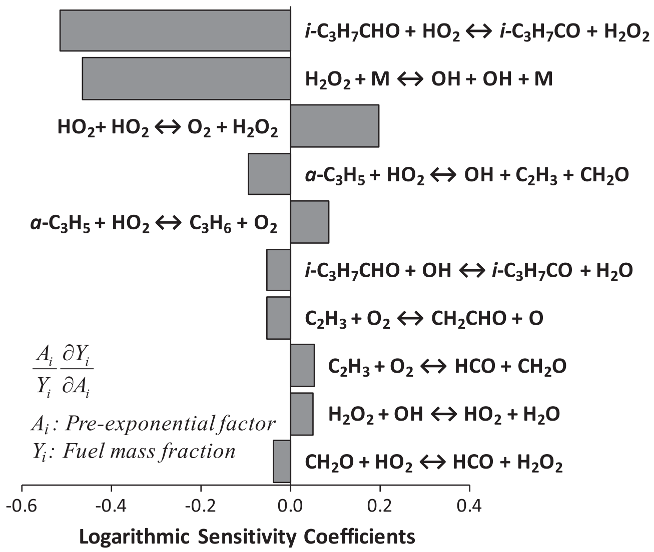
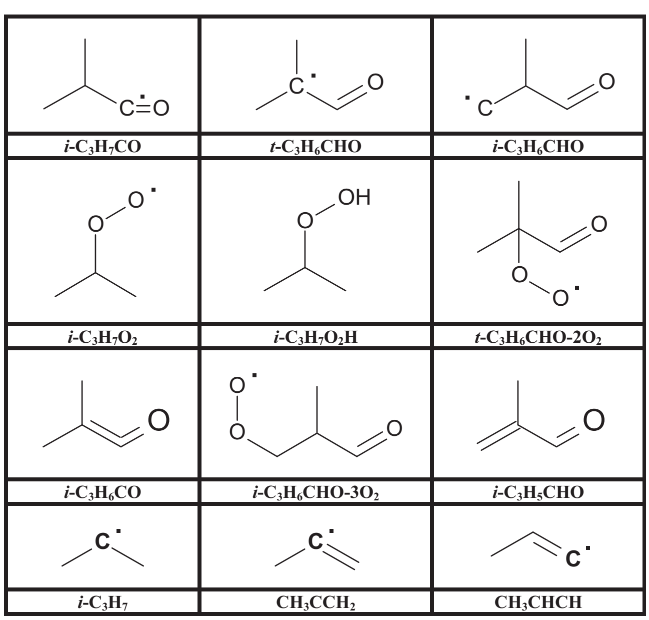


Fig. 14. First-order sensitivity analysis of the oxidation of *iso*-butanal at T = 800 K, φ = 1.0, τ = 0.7 s, and p = 10 atm, at approximately 60% of the total fuel conversion.

Table 2
Structure and nomenclature of selected species relevant to the oxidation of *iso*-butanal.



case, these results are consistent with the experimental observations of Salooja [49].

4.2. Laminar flame speeds

4.2.1. n-Butanal/air flames

The experimental and computed S_u^o's for n-butanal/air mixtures are depicted in Fig. 16, for $0.75 \leqslant \phi \leqslant 1.60$, T_u = 343 K, and p = 1 atm. The peak S_u^o of 52 cm/s was measured at ϕ = 1.10. The predicted values are in excellent agreement with the data for all ϕ's considered. Integrated reaction pathway flux analysis was performed for fuel-lean, stoichiometric, and fuel-rich n-butanal/air freely propagating flames. Since the results for fuel-lean and stoichiometry were nearly identical, only stoichiometric and fuel-rich (ϕ = 1.70) pathways are shown in Fig. 17. The primary difference in the decomposition pathways between stoichiometric and fuel-rich flames is in the branching ratios for the initial fuel decomposition reactions. Subsequent intermediates formation and consumption pathways are unchanged between fuel-lean to fuel-rich conditions. At ϕ = 1.0, fuel decomposition is dominated by H-abstraction reactions with H. The branching ratios of initial fuel specific radicals are 30:30:30:10 producing n-C_3H_7CO, C_3H_6CHO-2, C_3H_6CHO-3, and C_3H_6CHO-4, respectively. At ϕ = 1.70, the above ratios shift slightly to 25:20:20:5 and there is a notable increase in fuel consumption via the unimolecular pathways:

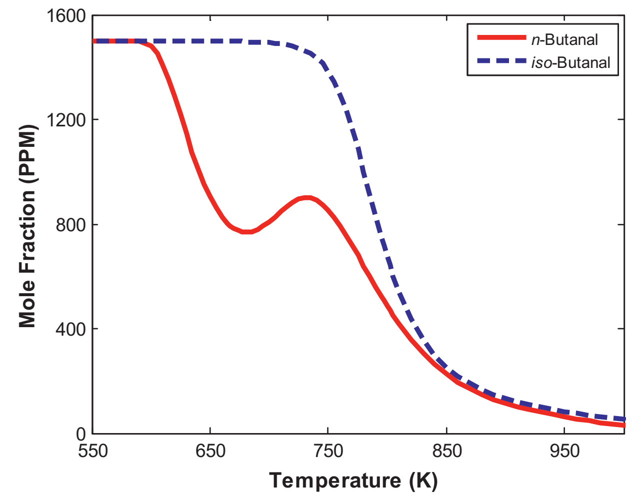


Fig. 15. Comparison of n-butanal and *iso*-butanal concentration profiles at τ = 0.7 s, p = 10 atm, and ϕ = 1.0.

$$n\text{-}C_3H_7CHO \leftrightarrow C_2H_5 + CH_2CHO \quad \text{(R21)}$$

$$n\text{-}C_3H_7CHO \leftrightarrow n\text{-}C_3H_7 + HCO. \quad \text{(R22)}$$

R21 and R22 account for approximately 20% of the total fuel decomposition at fuel-rich conditions. At flame temperatures, the decomposition of these fuel specific radical species occurs exclusively through β-scission reactions. C_3H_6CHO-4 decomposition produces C_2H_4 and CH_2CHO:

$$C_3H_6CHO\text{-}4 \leftrightarrow C_2H_4 + CH_2CHO. \quad \text{(R23)}$$

C_3H_6CHO-3 decomposition produces C_3H_6 and HCO:

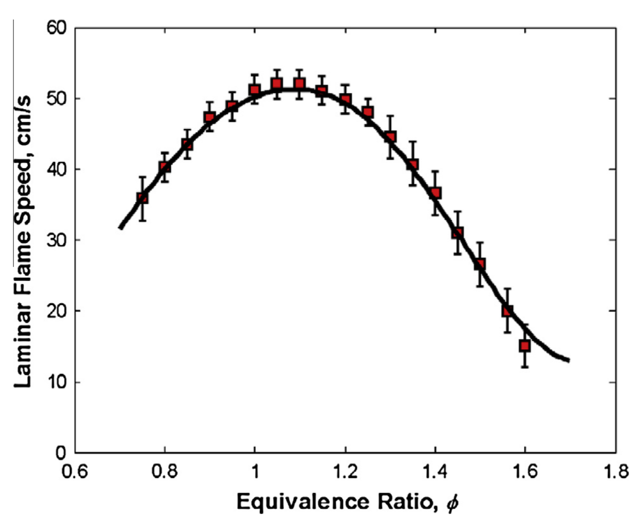


Fig. 16. Experimental and computed laminar flame speeds of n-butanal/air flames at T_u = 343 K and p = 1 atm. Symbols: experimental data; Lines: simulation results.

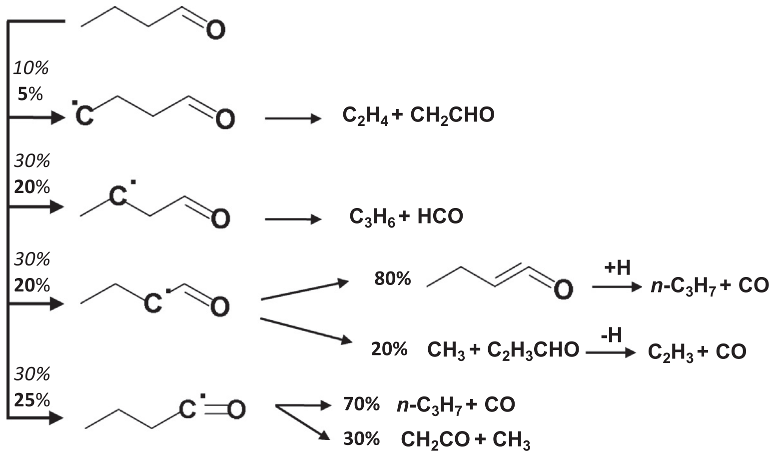


Fig. 17. Reaction path analysis for a ϕ = 1.0 (italicized text) and 1.7 (bold text) freely propagating n-butanal/air flames computed using the present model. The numbers indicate the conversion percentage.

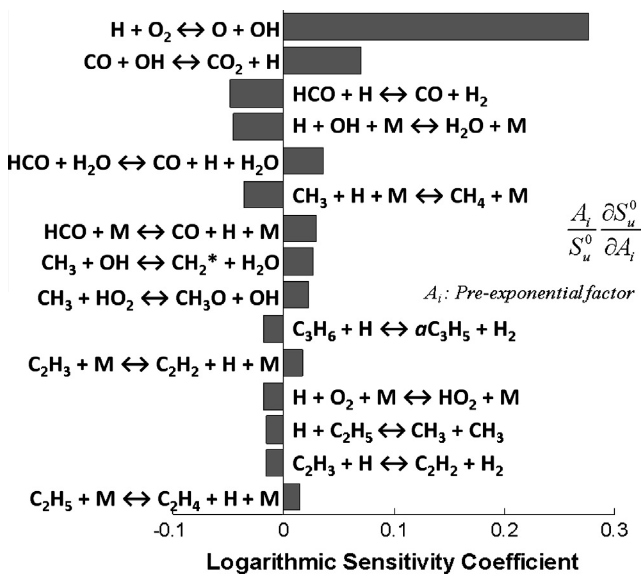


Fig. 18. Logarithmic sensitivity coefficients of laminar flame speed to kinetics for an n-butanal/air flame at ϕ = 1.0 and T_u = 343 K.

$$C_3H_6CHO\text{-}3 \leftrightarrow C_3H_6 + HCO. \quad (R24)$$

C_3H_6CHO-2 decomposes via two pathways:

$$C_3H_6CHO\text{-}2 \leftrightarrow C_2H_5CHCO + H \quad (R25)$$
$$C_3H_6CHO\text{-}2 \leftrightarrow C_2H_3CHO + CH_3. \quad (R26)$$

R25, resulting in the production of ethylketene (C_2H_5CHCO) and H, is favored over R26 which results in CH_3 and C_2H_3CHO; the R25:R26 reaction rate ratio is 8:2. In the present model ethylketene (C_2H_5CHCO) reacts with either OH or H resulting in n-C_3H_7 and CO_2 or CO respectively:

$$C_2H_5CHCO + OH \leftrightarrow n\text{-}C_3H_7 + CO_2 \quad (R27)$$
$$C_2H_5CHCO + H \leftrightarrow n\text{-}C_3H_7 + CO. \quad (R28)$$

It is more than likely that the reaction pathways for C_2H_5CHCO are incomplete, but since, as it will be shown next, there is no sensitivity of S_u^o's to large molecular weight stable oxygenated species chemistry, this is not an issue. n-C_3H_7CO has two decomposition pathways:

$$n\text{-}C_3H_7CO \leftrightarrow n\text{-}C_3H_7 + CO \quad (R29)$$
$$n\text{-}C_3H_7CO \leftrightarrow CH_2CO + CH_3 \quad (R30)$$

with the reaction rate ratio of R29 to R30 being 7:3 in the present model. Sensitivity of S_u^o's to rate constants of a ϕ = 1.0 n-butanal/air flame is shown in Fig. 18. As expected, results show that high-temperature oxidation chemistry of n-butanal is dominated by the reaction kinetics of H_2, CO, C_1, and C_2 species. Initiation reactions and reactions involving primary intermediates of fuel breakdown exhibit minor sensitivities.

4.2.2. iso-Butanal/air flames

The experimental and computed S_u^o's for *iso*-butanal/air flames are shown in Fig. 19, for $0.75 \leqslant \phi \leqslant 1.50$, T_u = 343 K, and p = 1 atm. The peak S_u^o of 48.8 cm/s was measured at ϕ = 1.10. The computed values using the present model are in excellent agreement with the data for *iso*-butanal/air flames as well. Although the low to intermediate temperature sub-model is not performing particularly well for *iso*-butanal oxidation in the JSR studies, the high-temperature oxidation mechanism for *iso*-butanal seems to be well represented by the present model. Figure 20 depicts the integrated reaction pathway flux analysis for ϕ = 1.0 and fuel-rich ϕ = 1.70 freely propagating *iso*-butanal/air flames. Analysis results for fuel-lean flames were almost identical to the ϕ = 1.0 case, hence are not presented herein. For the ϕ = 1.0 flame, initiation is dominated by H-abstraction. The branching ratios at ϕ = 1.0 are 30:15:55 for the production of i-C_3H_6CHO, t-C_3H_6CHO, and i-C_3H_7CO. There is increased consumption of *iso*-butanal, at fuel-rich conditions, through the unimolecular decomposition pathway:

$$i\text{-}C_3H_7CHO \leftrightarrow i\text{-}C_3H_7 + HCO. \quad (R31)$$

Subsequently, the branching ratio at ϕ = 1.70 for the production of i-C_3H_6CHO, t-C_3H_6CHO, i-C_3H_7CO, and i-C_3H_7 is 52:22:18:8. β-Scission reactions dominate the subsequent reaction pathways for the fuel specific radical species, i.e.:

$$i\text{-}C_3H_7CO \leftrightarrow i\text{-}C_3H_7 + CO \quad (R32)$$

$$i\text{-}C_3H_6CHO \leftrightarrow C_3H_6 + HCO \quad (R33)$$

$$i\text{-}C_3H_6CHO \leftrightarrow CH_3 + C_2H_3CHO \quad (R34)$$

$$t\text{-}C_3H_6CHO \leftrightarrow i\text{-}C_3H_5CHO + H \quad (R35)$$

$$t\text{-}C_3H_6CHO \leftrightarrow i\text{-}C_3H_6CO + H. \quad (R36)$$

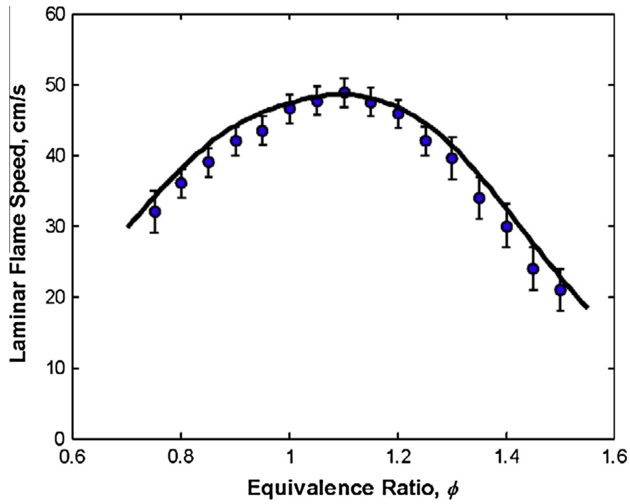


Fig. 19. Experimental and computed laminar flame speeds of *iso*-butanal/air flames at T_u = 343 K and p = 1 atm. Symbols: experimental data; Lines: simulation results.

i-C_3H_6CHO has two decomposition pathways with the path producing C_3H_6 and HCO (R33) favored over the production of CH_3 and C_2H_3CHO (R34). *t*-C_3H_6CHO reacts and produces either methlypropenal (*i*-C_3H_5CHO) or dimethylketene (*i*-C_3H_6CO). *i*-C_3H_5CHO decomposition pathway is via H-abstraction from the formyl group forming CH_3CCH_2 and CO. *i*-C_3H_6CO reacts with OH via:

$$i\text{-}C_3H_6CO + OH \leftrightarrow i\text{-}C_3H_7 + CO_2. \qquad \text{(R37)}$$

In the present model, the oxidation pathways for these two intermediates, i.e. *i*-C_3H_5CHO and *i*-C_3H_6CO, are not complete. Nevertheless as stated above, this is not expected to affect the computed results given that the flame behavior is not sensitive to the kinetics of large oxygenated stable intermediates.

The logarithmic sensitivity coefficients of S_u^o to kinetics were computed using the present model and depicted in Fig. 21 for a ϕ = 1.0 freely propagating *iso*-butanal/air flame. The results shown in Fig. 21 are very similar to those reported earlier for the *n*-butanal/air flame with a few notable exceptions. *iso*-Butanal/air flames have a pronounced negative sensitivity to reactions involving C_3 hydrocarbons, namely:

$$C_3H_6 + H \leftrightarrow a\text{-}C_3H_5 + H_2 \qquad \text{(R38)}$$
$$C_3H_6 + OH \leftrightarrow a\text{-}C_3H_5 + H_2O \qquad \text{(R39)}$$
$$a\text{-}C_3H_5 + H + M \leftrightarrow C_3H_6 + M \qquad \text{(R40)}$$

iso-Butanal/air flames have a pronounced propensity to produce C_3H_6 relative to say, *n*-butanal/air flames. The consequences of this statement will be discussed in detail next when comparing flames of these two C_4 aldehydes.

4.2.3. Comparison of laminar flame speeds

In Fig. 22, the experimental and computed S_u^o's of *n*- and *iso*-butanal/air flames are compared. Over all ϕ's considered both experimental and computed results indicate that *n*-butanal/air flames propagate faster relative to *iso*-butanal/air flames. To be more precise, *n*-butanal/air flames propagate between 15% and 30% faster than *iso*-butanal/air flames, with this percentage increasing with ϕ. This is expected as CH_3 substitution (i.e., branching) in the fuel molecular structure has long established (e.g., [75]) to result in resonantly stable intermediates reducing overall reactivity and therefore S_u^o. The primary cause for this difference is the different C_3H_7 radical produced in each flame. *n*-Butanal produces readily *n*-C_3H_7 while *iso*-butanal produces *iso*-propyl radicals (*i*-C_3H_7). At flame temperatures, *n*-C_3H_7 and *i*-C_3H_7 decompose via the following β-scission reactions:

$$n\text{-}C_3H_7 \leftrightarrow C_2H_4 + CH_3 \qquad \text{(R41)}$$
$$i\text{-}C_3H_7 \leftrightarrow C_3H_6 + H. \qquad \text{(R42)}$$

Figure 23 depicts the relative concentrations of C_3H_6 and C_2H_4 in fuel-rich flames of *n*- and *iso*-butanal. Clearly there is a higher concentration of C_2H_4 and lower concentration of C_3H_6 in the *n*-butanal/air flame and the reverse is true for the *iso*-butanal/air flame. Although R42 produces the highly reactive H radical, C_3H_6 consumes H and results in the somewhat unreactive, resonantly stable allyl radical (*a*-C_3H_5) via

$$C_3H_6 + H \leftrightarrow a\text{-}C_3H_5 + H_2 \qquad \text{(R43)}$$
$$a\text{-}C_3H_5 + H + M \leftrightarrow C_3H_6 + M \qquad \text{(R44)}$$

reducing thus the overall reactivity and as a result S_u^o for the *iso*-butanal/air flame.

Figure 24 depicts a comparison between S_u^o's of *n*-butanal/air and propanal/air flames; S_u^o's for propanal/air flames have been reported in a recent study by the authors [55]. For $0.75 \leqslant \phi \leqslant 1.20$, there is no difference between the experimental S_u^o's of flames of

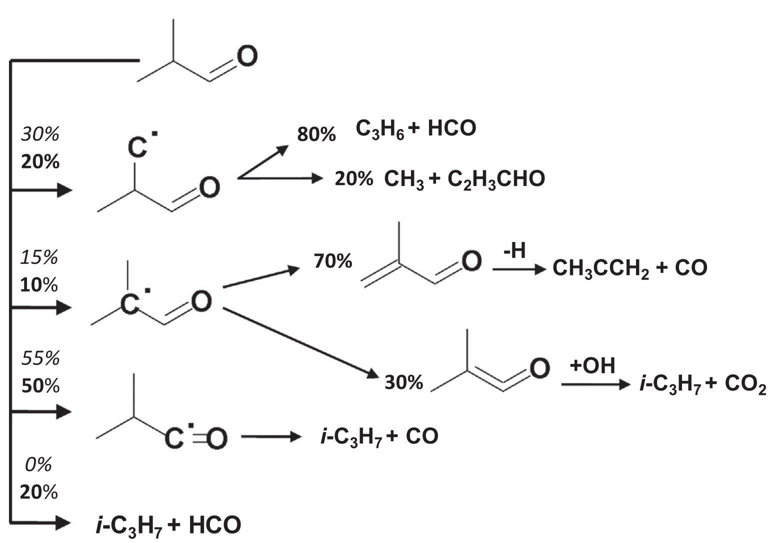


Fig. 20. Reaction path analysis for a ϕ = 1.0 (italicized text) and 1.7 (bold text) freely propagating *iso*-butanal/air flames computed using the present model. The numbers indicate the conversion percentage.

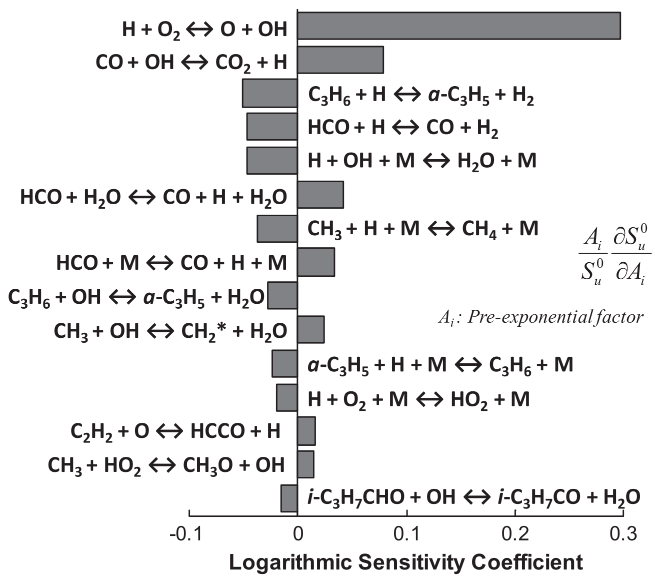


Fig. 21. Logarithmic sensitivity coefficients of laminar flame speeds to kinetics for an *iso*-butanal/air flame at $\phi = 1.0$ and $T_u = 343$ K.

these two fuels. For $\phi > 1.20$, the experimental results indicate that propanal/air flames are between 6% and 50% faster than *n*-butanal/air flames. This trend is mirrored to a large extent in the numerical simulations using the present model, which however overpredicts slightly the reactivity of propanal/air flames at fuel-lean conditions. It is therefore somewhat suitable to use the present model to gain further insight into the kinetic reason for the observed difference.

In an attempt to understand the change that occurs at fuel-rich conditions, the computed OH and HCO profiles for propanal/air and *n*-butanal/air flames at $\phi = 0.9$ and 1.60 are compared in Figs. 25 and 26 respectively, and various trends become immediately apparent. As expected, there is an order of magnitude reduction in the peak concentrations of OH and HCO at fuel-rich conditions in both flames, this causes the increase in fuel consumption pathways through unimolecular decomposition observed in Figs. 17 and 20. Furthermore, the OH and HCO concentrations for the *n*-butanal/air flame are notably lower compared to the propanal/air flame, resulting thus in higher overall reactivity and S_u^o's for propanal flames. There is a number of kinetic pathways that account for this trend, but the analysis of the present model tends to indicate that it is a direct result of the intermediates produced via thermal decomposition of propanal and *n*-butanal. At fuel-rich

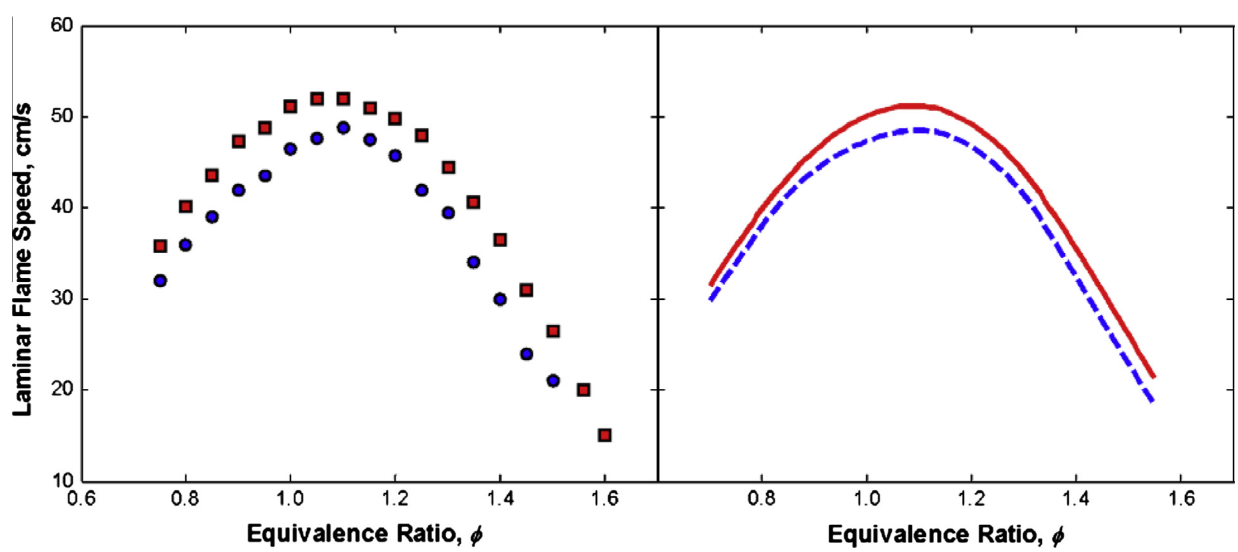


Fig. 22. Comparison of experimentally determined and numerically computed laminar flame speeds of *n*-butanal/air (□ and ▬▬▬) and *iso*-butanal/air (○ and ▬ ▬ ▬) flames at $T_u = 343$ K and $p = 1$ atm.

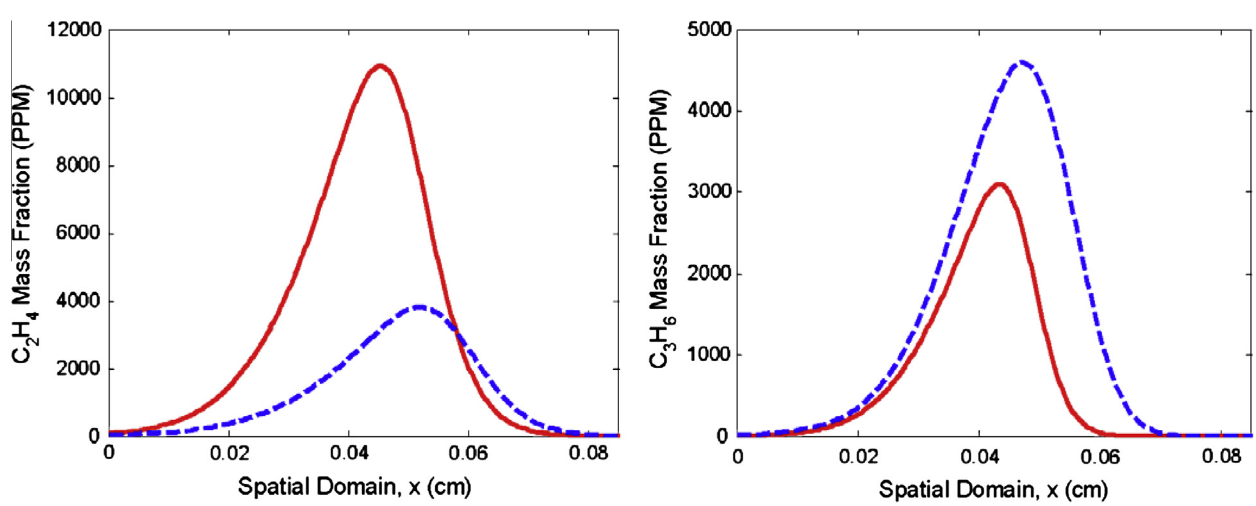


Fig. 23. Computed C_2H_4 and C_3H_6 mass fraction profiles for *n*-butanal/air (solid lines) and *iso*-butanal/air (dashed lines) flames at $\phi = 1.4$, $T_u = 343$ K, and $p = 1$ atm.

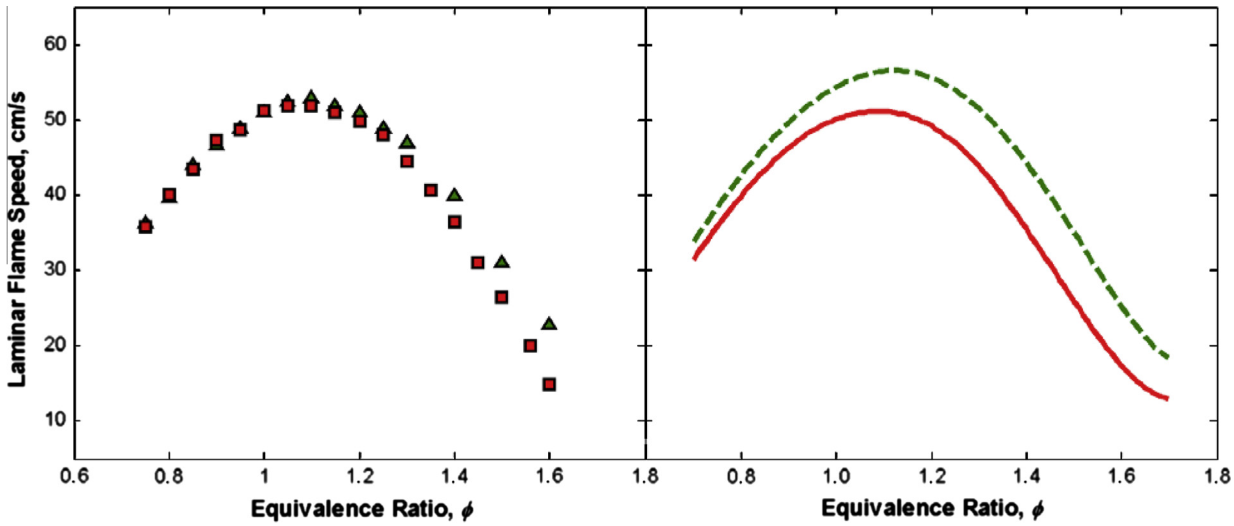


Fig. 24. Comparison of experimentally determined and numerically computed laminar flame speeds of *n*-butanal/air (□ and ▬▬) (present study) and propanal/air (△ and ▬ ▬ ▬) (from Ref. [55]) flames at T_u = 343 K and p = 1 atm.

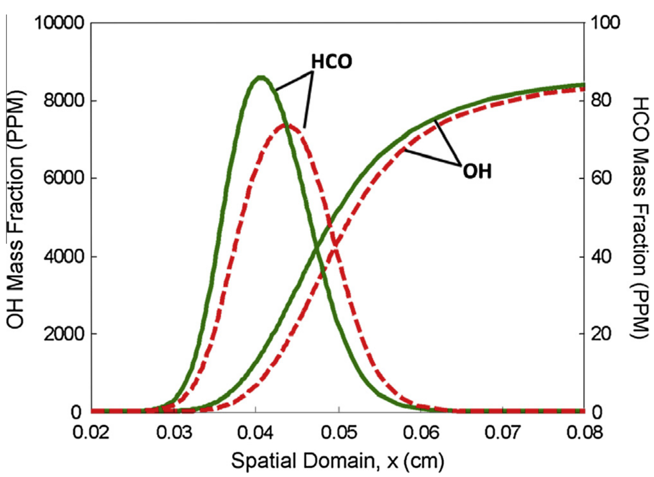


Fig. 25. Computed HCO and OH mass fraction profiles for propanal/air (solid lines) and *n*-butanal/air (dashed lines) flames at ϕ = 0.9, T_u = 343 K, and p = 1 atm.

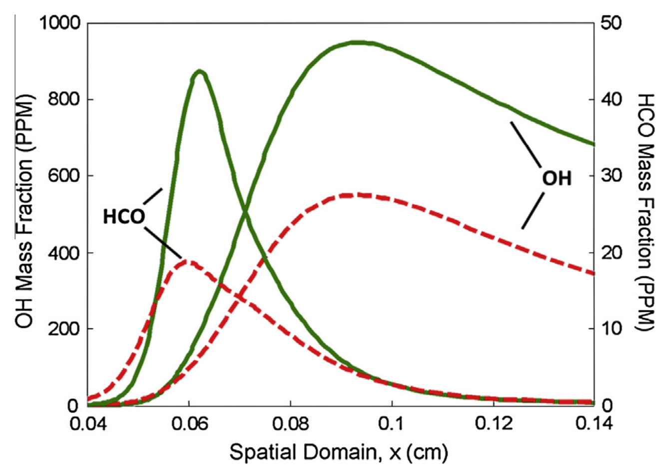


Fig. 26. Computed HCO and OH mass fraction profiles for propanal/air (solid lines) and *n*-butanal/air (dashed lines) flames at ϕ = 1.60, T_u = 343 K, and p = 1 atm.

conditions, *n*-butanal decomposes via R21 and R22 resulting in C_2H_5, CH_2CHO, *n*-C_3H_7, and HCO. At fuel-rich conditions, propanal undergoes unimolecular decomposition via

$$n\text{-}C_2H_5CHO \leftrightarrow C_2H_5 + HCO \tag{R45}$$
$$n\text{-}C_2H_5CHO \leftrightarrow C_2H_4 + CH_2O. \tag{R46}$$

For a ϕ = 1.60 propanal/air flame, the largest fraction of HCO is produced via

$$CH_2O + H \leftrightarrow HCO + H, \tag{R47}$$

with CH_2O being produced directly from R46. Decomposition of CH_2CHO and *n*-C_3H_7 in *n*-butanal/air flames results in additional pathways for the production of CH_3. At fuel-rich conditions, both *n*-butanal and propanal flames exhibit additional pathways to radical formation and consumption. These pathways result in a more reactive radical pool for propanal decomposition compared to *n*-butanal decomposition.

5. Concluding remarks

A comprehensive experimental and chemical kinetic modeling study of the oxidation of *n*-butanal and *iso*-butanal, both saturated C_4 aldehydes, was performed. The evolution of reactants, products, and intermediates were measured during the oxidation of these aldehydes in a jet stirred reactor at 10 atm, from equivalence ratios ranging from 0.3 to 2.0, and reactor temperatures ranging from 500 K to 1100 K. Laminar flame speeds of *n*- and *iso*-butanal/air flames were determined in the counterflow configuration at atmospheric pressure and for an unreacted mixture temperature of 343 K. The experimental data were used to develop and validate a detailed, interpretive chemical kinetic reaction model consisting of 244 species and 1198 elementary reactions.

Considering first the jet stirred reactor validations, the present chemical model was able to reproduce closely detailed concentration profiles for all species considered during *n*-butanal oxidation. Regarding the *iso*-butanal oxidation, the present model reproduced the major species concentration profiles accurately. Nevertheless deficiencies were found in the present model's ability to reproduce experimental results for minor intermediates. It is suggested that an incorrect distribution of radical isomers of C_3H_5 following the decomposition of propene at jet stirred reactor relevant conditions might be responsible for this discrepancy. However, using the present kinetic model the laminar flame speeds were predicted closely.

Detailed sensitivity and reaction pathway analyses were performed. The results revealed that the jet stirred reactor and flame data are sensitive to unique subsets of the kinetic model providing thus a highly constraining set of validation targets.

The jet stirred reactor results were used to compare the relative reactivities of the *n*-butanal, *iso*-butanal, and previously determined results for propanal over the low-temperature reactivity regime. *iso*-Butanal was determined to be the least reactive, demonstrating little or no low-temperature chemistry and *n*-butanal the most reactive. The reactivity of propanal lies between the low-temperature reactivities of *n*- and *iso*-butanal. Similarly, *n*-butanal/air flames were determined to propagate faster than *iso*-butanal/air flames. Interestingly, both experimental and computed results revealed that fuel-rich propanal flames propagate faster than fuel-rich *n*-butanal flames. Detailed analysis was performed using the present chemical kinetic reaction model to provide insight into all aforementioned observations.

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Appendix A. Supplementary material

This publication includes the following supplemental material:

1. An interpretive chemical kinetic model (244 species and 1198 reactions) covering high and low temperature reactivity of n-butanal, iso-butanal, and propanal in CHEMKIN format.
2. A smaller interpretive chemical kinetic model covering the high temperature reactivity of n-butanal, iso-butanal, and propanal in CHEMKIN format used for flame speed calculations.
3. The proposed thermodynamic datafile in CHEMKIN format.
4. The proposed transport datafile in CHEMKIN format.
5. Experimental data for laminar flame speeds of n-butanal/air, iso-butanal/air, and n-propanal/air flames (in EXCEL format).
6. Experimental JSR data for the oxidation of n-butanal/oxidizer, iso-butanal/oxidizer, and propanal/oxidizer mixtures (in EXCELformat).

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.combustflame.2013.03.018.

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